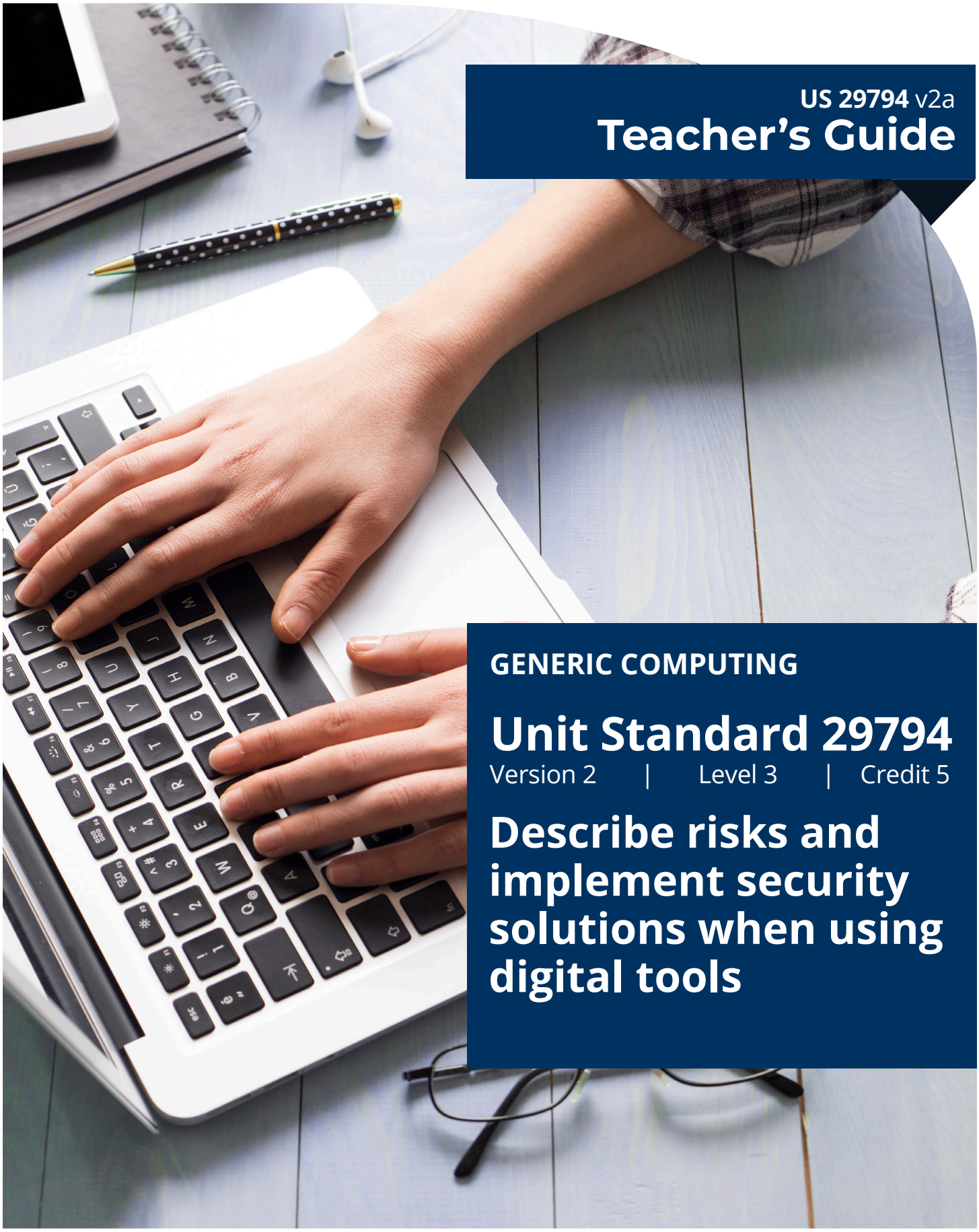


US 29794 v2a

Teacher's Guide

GENERIC COMPUTING

Unit Standard 29794

Version 2 | Level 3 | Credit 5

Describe risks and implement security solutions when using digital tools

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NZQA Information

The information contained in this resource is relevant to assessment against the following NZQA unit standard from the Computing – Generic Computing domain.

U/S	LEVEL	CREDIT	TITLE
29794v2	3	5	Describe risks and implement security solutions when using digital tools

The information covered in the resource may also relate to unit standards in other domains.

Level Descriptor: The unit standard is at level 3 of the NQF. At this level, the following is expected of the learner:

KNOWLEDGE	SKILLS	APPLICATION [OF KNOWLEDGE AND SKILLS]
Some operational and theoretical knowledge in a field of work or study	Select and apply from a range of known solutions to familiar problems Apply a range of standard processes relevant to the field of work or study	Limited supervision Requiring major responsibility for own learning and performance Adapting own behaviour when interacting with others Contributing to group performance

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instant LTD education solutions

21 Ruakura Rd, Hamilton East,
Hamilton 3216, New Zealand
Phone: +64 800 864 863
E-mail: orders@instant.org.nz

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About this Teacher's Guide

Learning Purpose & Outcomes

The purpose of this guide is for learners to be able to demonstrate their understanding and ability to apply compliance requirements when working with digital tools and information.

Learners who successfully complete this course will be able to do the following in an organisation contex:.

- Identify and describe risks and compliance requirements when using digital tools, and digitally stored and transmitted information.
- Explain procedures and implement solutions to meet security and good practice requirements when using digital tools.

A glossary containing difficult or technical terms has been provided at the end of this guide. These words are highlighted in the main text.

Unless stated otherwise, any organisations or individuals referred to in the examples and activities in this guide are fictitious. Any resemblance to actual organisations or persons is coincidental.

General Information

The teaching and learning guides are designed to prepare learners for the assessment (of the identified NZQA standard).

1. This Teacher's Guide is provided for use by teachers only. It is not intended that learners will have access to this document. However, it is expected that, where relevant, teachers will provide learners with suggested answers from this document.
2. These guides are not intended as stand alone distance-learning or self-directed study packages. Teachers should add their own learning resources and activities as appropriate to ensure a robust learning pedagogy. The pedagogy engaged should be suitable to the particular organisation or school the teacher is employed in. This includes appropriate extension learning as required.
3. Learners may need additional literacy and numeracy support as they work through this course. The teacher needs to ensure that this support is in place for students.
4. Website and application (including apps) content changes regularly. This may mean that changes have been made to content that is mentioned in the Learner's Guide and the Teacher's Guide. It is important that any links or applications are previewed to ensure they are active or current, before giving them to students.
5. Laws and compliance regulations are regularly amended and/or repealed. Teachers should research and incorporate any recent legislative or compliance changes where required.

Teacher's Note:

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LESSON 1:

Privacy and security – Compliance requirements and risks

Learning Objectives

When you have finished this lesson, you will be able to identify and describe the following as it relates to the use of digital tools:

- Legal and regulatory risks and compliance requirements related to privacy of information.
- The implications of these risks and compliance requirements for data, systems, and organisational processes.

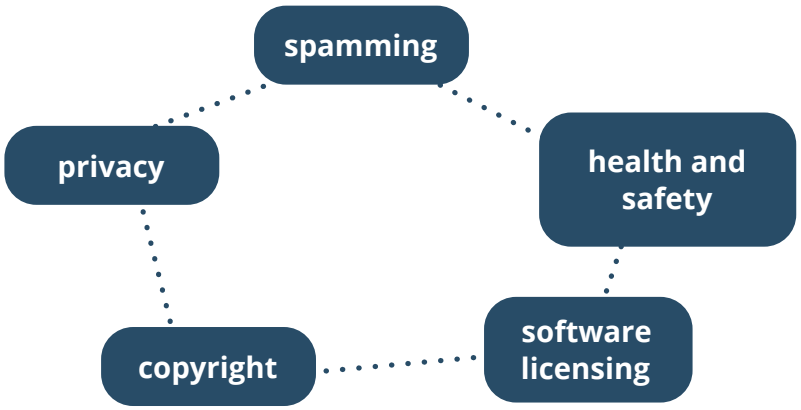
A space has been left on the right of every page for you to make notes about what you are learning.

Introduction

In this course, we look at how to deal with a range of risks when using digital tools in an **organisation** context.

An organisation can be a business, place of learning (eg school, training institution), volunteer organisation (eg charity), etc.

This includes risk related to the following.



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Many of these risks relate to legal and regulatory **compliance requirements**.

Compliance requirement	A law, rule, or standard that must be met.

The purpose of these compliance requirements is to help protect organisations, individuals, and property when it comes to:

- using digital devices
- transmitting (sharing) and storing digital information.

Regardless of the type of organisation you work for or study at, there will be several compliance requirements that you must adhere to when you work with digital tools and information. If you do not comply with these requirements, you and/or the organisation could risk getting a fine or other type of legal sanction.

Legal sanction	A punishment under the law (eg a fine, being jailed, etc).

For each of the areas of potential risk, we will look at the:

- compliance requirements (laws and regulations) relevant to each
- implications for organisations in terms of how they deal with data, the systems they have in place, and organisational processes
- risks faced by organisations if they do not meet the compliance requirements.

We will also look at the important role an organisation's policies and procedures play in compliance and addressing risks.

In this lesson, we begin by looking at the compliance requirements and risks related to the privacy and security of digital information.

Keeping data secure and private

Organisations store a lot of data about their business which they need to keep secure. The type of data may include:

- **personal information** about individuals, eg staff home address, telephone numbers; customer details; etc.
- Marketing strategies
- Product development information
- Financial information

In this lesson we will focus on the privacy of personal information. However security of all digital information is important to an organisation.

Personal information	Anything that can identify a person. For example, a photograph of the person, their name, address, etc.

https://privacy.org.nz/assets/New-order/Your-responsibilities/Privacy-resources-for-organisations/2020-A-quick-tour-of-the-privacy-principles-Oct-2020.pdf

Everyone has the right to the privacy of their personal information under NZ law. The legislation that sets the rules for the collection, storage and release of personal information is the **Privacy Act 2020**. The aim of the law is to give individuals greater control over their personal information.

The Privacy Act lists thirteen **Privacy Principles**.

Privacy Principles

1. Collection of information must be for a lawful reason.
2. If possible, you should collect the information directly from the individual person.
3. The individual should be made aware that their information is being collected.
4. The information should not be collected in an intrusive manner.
5. The information collected must be securely stored.
6. The person has the right to access information you have about them.
7. The person has the right to correct information you have about them.
8. You need to take all reasonable steps to ensure the information is accurate.
9. Personal information should not be kept longer than needed (for the purpose it was collected).
10. Information obtained for one purpose should not be used for another purpose without the consent of the individual.
11. Personal information should not be shared with others without good reason.
12. Personal information can only be sent overseas if the information will be adequately protected or with the person's permission.
13. A unique identifier (instead of the person's name) should only be used if needed.

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For organisations that store personal information, there are several risks related to privacy compliance requirements. We are going to focus on those related to information being:

- securely stored
- kept accurate.

Security: Personal information must be kept secure and unauthorised access to the information prevented (Principles 5 and 11). This will be discussed in the section on **Privacy (data) breach**.

Accuracy: Information must be accurate. If the organisation becomes aware that some of the information is inaccurate, all reasonable steps must be taken to correct it (Principles 7 and 8). This will be discussed in the section on **Data corruption**.

Loss of data

People share their personal information (eg address, credit card details) with organisations and expect these organisations to keep the information private. A data breach happens when someone's personal information is accessed by, or given to, an unauthorised person (ie the information is stolen or shared by mistake).

Data can also be lost due to other causes such as a natural disaster. Let's have a look at how a data breach or data loss could happen.

Activity

If you send an email to several recipients, with all recipients' email addresses in the **To:** field, you may be revealing personal information.

What personal information are you revealing?

People's email addresses (and possibly also their names).

Why could this be a problem?

Someone else could use these email addresses to send people unsolicited information (eg advertisements), or to contact them. [Any other suitable answer]

Malware attack

You may have heard of computer viruses, but there are several other forms of malware (malicious software) that can be easily transmitted in a digital environment.

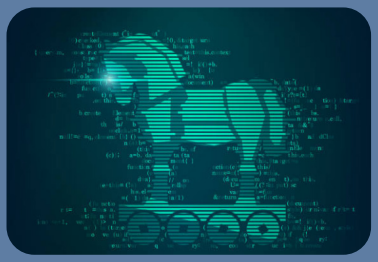


Viruses

A computer virus is a self-replicating piece of software that inserts itself into legitimate computer programs. Just like a biological virus, computer viruses can be dormant for a long time and then suddenly replicate with great speed. Some viruses are written with the intent to destroy computer files or the drive so data can never be retrieved.

Trojans

A Trojan is a piece of software that appears to do one thing but actually does another. It may contain a virus or code that enables others to access your computer or it copies your keystrokes. This lets others get access to your passwords and personal details





Worms

Worms, unlike viruses, do not need to attach themselves to an existing program. They use a lot of your computer's programming capacity and slow it down. Some worms only slow computers down, but others will delete files, or hold your computer to ransom. If your computer is held to ransom, it means you will not be able to access your computer or your files until you pay someone money to unlock it remotely

Natural disasters

Natural disasters include events such as fire, flood, earthquake, or tornado.

Your computer may be damaged, destroyed or unable to be retrieved in a disaster.



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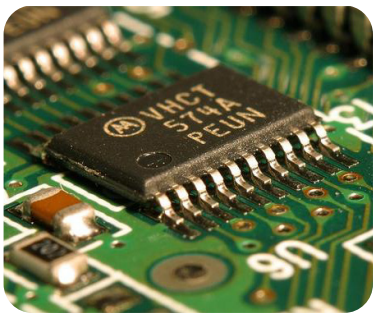
Power surge or outage



The power always goes off just when you don't want it to! In a power cut or unexpected outage, any work that has not been saved will be lost.

Electricity supply can also surge (or spike) when you turn a computer on or off, or when there is a surge on the electricity grid (eg caused by a lightning strike). Surges in electricity can damage the microprocessors, hard disks, and circuits.

Hardware failure



Hardware refers to the electronics and components in a computer. This includes the storage drives, motherboard, and memory. It also includes peripherals such as your monitor, mouse, and keyboard.

If your computer hardware fails, it could make it impossible to retrieve the information stored on your computer.

Data corruption

Data corruption refers to unintended changes made to original data when it is stored, processed, transmitted, etc. Data corruption could be because of:

- **malicious intent** – someone who sets out to alter the data on purpose to do damage (eg using malware)
- **unexpected program exits** – eg caused by a power failure or power surge, or by your computer 'crashing'
- **hardware defects** – eg a faulty hard drive
- **human error** – people incorrectly entering, changing, or deleting data.

With regards to inaccurate information, the Privacy Act states that organisations are responsible for taking all reasonable steps to ensure personal information they hold is accurate, complete, and up to date.

Unauthorised access

Unauthorised access is when someone who has not been given permission is able to access data. There are several intentional and unintentional reasons why someone may be able to gain unauthorised access

- **Data interception:** When data is transferred (sent) over a network, it is vulnerable to being accessed, changed, or stolen.
- **Poor security:** The data is not properly secured (eg password protected), people share their login details with others, etc.
- **Hacking:** Someone is able to get around the security that is in place.
- **Human error:** Someone is sent the data by mistake. For example, you put the wrong email address in the To: field.
- **Incorrect disposal of hardware:** Throwing away, recycling, or giving someone else your device without first fully deleting all data.
- **Intentional data sharing:** Intentionally sharing private information with someone who is not authorised to have that information. For example, someone (who is not authorised to have this information) asks for an employee's personal email address or phone number.
- **Theft:** Someone copying data from a device (without permission).
- **Fraud:** Someone pretending to be someone else and requesting information. For example, someone pretending to be a bank worker, asking for someone's online banking login details. This includes **phishing scams**.



Phishing scam	Phishing (pronounced 'fishing') is a way people try to get sensitive information (eg usernames, password, credit card details).

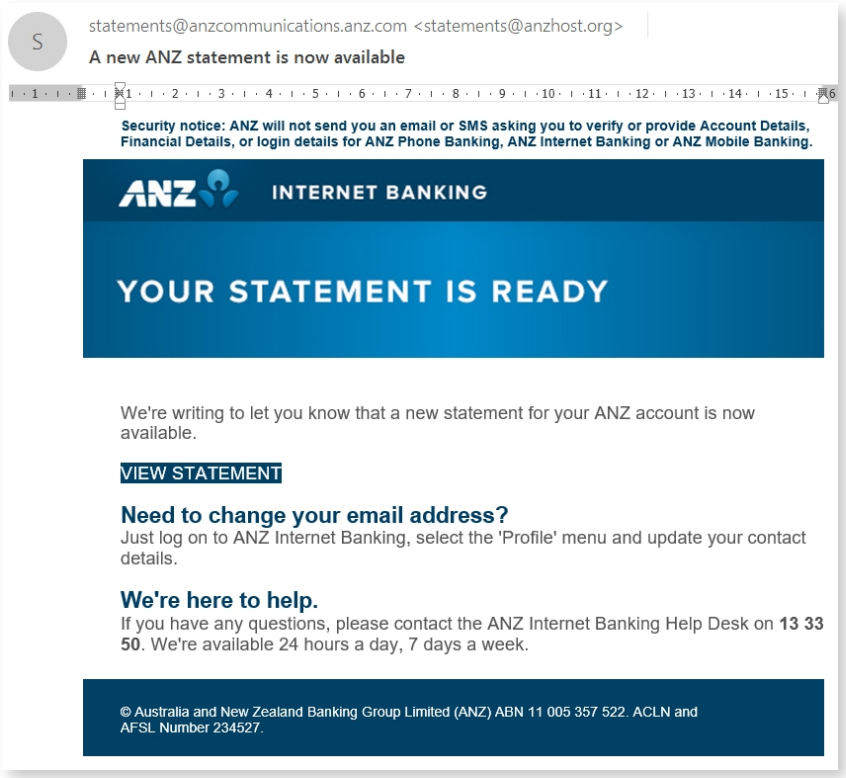
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Phishing scams

Phishing scams come in different forms, including email and web-based phishing scams. Two examples are shown below.

Email phishing scam example

Below is an example of a sophisticated email scam.



S

statements@anzcommunications.anz.com <statements@anzhost.org>

A new ANZ statement is now available

Security notice: ANZ will not send you an email or SMS asking you to verify or provide Account Details, Financial Details, or login details for ANZ Phone Banking, ANZ Internet Banking or ANZ Mobile Banking.

ANZ INTERNET BANKING

YOUR STATEMENT IS READY

We're writing to let you know that a new statement for your ANZ account is now available.

VIEW STATEMENT

Need to change your email address?
Just log on to ANZ Internet Banking, select the 'Profile' menu and update your contact details.

We're here to help.
If you have any questions, please contact the ANZ Internet Banking Help Desk on **13 33 50**. We're available 24 hours a day, 7 days a week.

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The email has been made to look like it has been sent by a well-known bank. It uses all the branding, contact details, etc of the bank, and it even has a security notice at the top of the email! However, there are some important clues that the email is a fraud.

- The sender's address is not a legitimate ANZ address.
- The person the email was sent to is not an ANZ customer, OR
  - they have not signed up to receive notifications that statements are available, OR
  - the email was sent to a different address than the one they used to sign up for statement notifications.
- If you **hover** your mouse over any link button, you may get additional clues that the email is a scam. **VERY IMPORTANT:** Be careful that you only **hover** over the link button. Do NOT click on it.



If you hover your mouse over the VIEW STATEMENT link shown in the email address above, you will see the following.

Continued on next page...

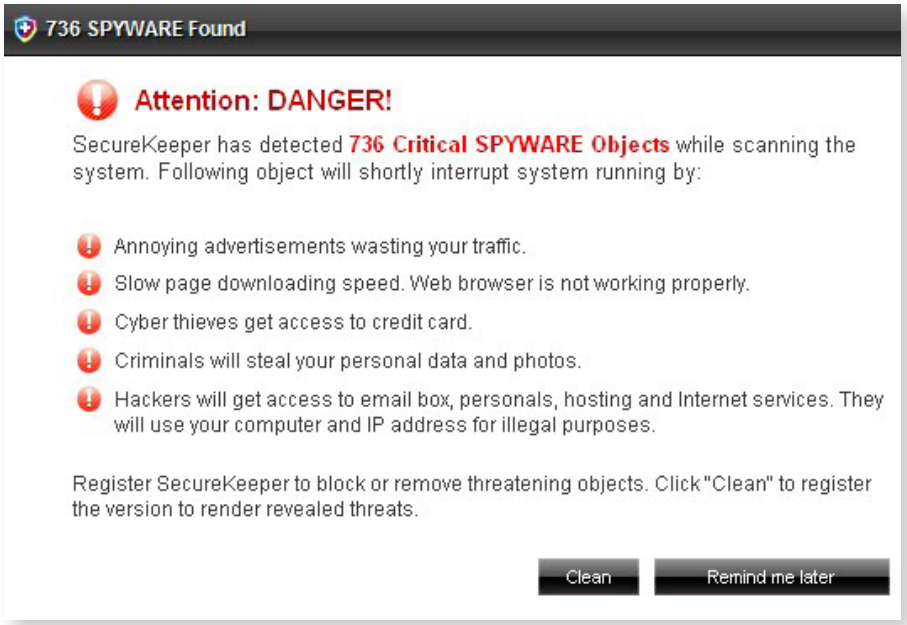
Notice that the link address is not a legitimate ANZ web address. In fact, there is no mention of ANZ at all in the link. It is for a SharePoint web address.

If you were to click on the link, there are several possibilities, including the following:

- You might be taken to a site that **looks** like it is a legitimate ANZ site, but is actually a 'fake' site, created to get you to input your ANZ internet account login details. The fraudster could then use this information to access your actual ANZ account.
- Malware could be downloaded onto your computer.

Internet phishing scam example

Below is an example of a potential phishing scam and/or virus attack. If you click on the 'Clean' button, malware may be downloaded onto your computer. If a message like this one does not come from your own anti-malware software, you know it is a virus attack or phishing scam.



- Do not press on any of the buttons!
- Close the window by pressing the small 'X' (usually at the top right corner of the pop-up).
- If you can't see a small 'X', close the browser tab or Window instead.

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Activity

The Department of Consumer Affairs has a website called Scamwatch (www.scamwatch.govt.nz). This site helps people identify when they are being scammed and gives advice on how to protect against scamming.

Have a look at the information on this site so that you can become more aware of the potential scamming threats currently on the Internet.

Network access

A network connection is two or more PCs which either connect through a server or router. They usually share files, devices (eg printers), and even an Internet connection. This is very useful in a work or school environment. It means that instead of each user needing to have their own printer, etc they can share one printer between them.

However this makes data vulnerable to the following:

- A virus infection on one computer will spread to others on the network.
- Unauthorised access to one computer may mean others can also be accessed.
- Sharing an internet connection increases the risk of someone accessing the network or a user downloading malware.



Risks

Below is a summary of the key risks to organisations that could result from a data breach and/or data corruption.



Identity theft

Identify theft is when someone accesses enough personal information that they can pretend to be that person. They may then use that person's identity for activities such as accessing loans in their name.

Access

The organisation may lose access to important data such as customer information, product information and financial records. Competitors may also then find out about commercially sensitive information.



Damage to an organisation's reputation

When data is accessed by an unauthorised person (either because of human error or criminal activity such as hacking), this can have serious implications for an organisation's reputation.

If someone's private information stored by an organisation becomes public knowledge, people may lose trust in the organisation. Let's look at the example of an ecommerce site.



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Example: Damage to an organisation's reputation

All ecommerce sites collect and store personal information such as their customers' names, addresses, credit card details, etc. If the site is hacked, customers will lose trust in the site and will be nervous about buying anything from them in future. They will also be likely to share their experience with their friends and family, and in this way, the company's reputation may be harmed. This, in turn could have negative financial implications for the organisation. If people do not trust the company and decide not to buy from the company in future, this can have a serious impact on their sales.

Implications for organisations

The legal requirements related to privacy mean that organisations need to have policies and procedures in place that ensure the:

- security of data
- accuracy of data.

If improper information is disclosed, this can have disastrous effects on the business and the individual(s) concerned (see the section on Risks).

Members of the public who feel that their privacy has been breached may make a complaint to the Privacy Commissioner. The Commissioner may decide to investigate the complaint. If they find that a breach has occurred, they are able to take Court proceedings for the breach. For example, someone whose privacy has been breached could ask for the organisation to compensate them for damages resulting from the breach (eg a financial settlement).

Organisations need to:

- think about what information they are transmitting via the Internet and make sure that it complies with the Privacy Act
- make sure that their computer systems are secure from potential threats
- ensure employees understand how and when to use information
- ensure employees are adequately trained to manage their own and others' data, and how to prevent data breaches
- have processes in place to manage any data breaches.

Organisations should include a Privacy Statement on their website and/or any communications they have with their customers/clients.

An example of a Privacy Statement is shown on the next page. It includes information about:

- the type of information being collected
- why it is collected
- the steps the organisation will take to ensure the security and privacy of the information
- how long the information will be held and how it will eventually be destroyed
- the fact that people have the right to ask for a copy of their personal information and/or to have it corrected.

The Privacy Statement on the following page was created using the automatic Privacy Statement Generator (Priv-o-matic), which can be found on the website of the Privacy Commissioner: https://www.privacy.org.nz in the Tools menu.



Privacy Statement Generator

Example: Privacy Statement

Privacy Statement

We collect personal information from you, including information about your:

- name
- contact information
- location
- computer or network
- billing or purchase information.

We collect your personal information in order to:

- provide you with excellent service and the best possible customer experience, tailored to your needs.

We keep your information safe by using encrypted storage and only allowing relevant client support staff to access the information.

We keep your information for seven years, at which point we securely destroy it by erasing all digitally held information.

You have the right to ask for a copy of any personal information we hold about you, and to ask for it to be corrected if you think it is wrong. If you'd like to ask for a copy of your information, or to have it corrected, please contact us at thecompany.co.nz, or 056 23151, or PO Box 8984, Luneville.

Type of info being collected.

Why the info is collected.

Steps the organisation will take to ensure the security and privacy of the info.

How long the info will be held and how it will be destroyed.

People have the right to ask for a copy of their personal info and/or to have it corrected.

The website also contains further, more detailed information about the Privacy Act and compliance requirements for organisations.

- Click on the **Your Responsibilities** menu item.

In the next lesson, we look in greater detail at the procedures an organisation can put in place to prevent and manage data breaches, and to ensure the data they hold is not corrupted (ie the accuracy of the data is maintained).

LESSON 2:

Other risks and compliance requirements

Learning Objectives

In previous lessons we looked at the risks and compliance requirements related to privacy, security, and health and safety. In this lesson, we look at some other key risks and compliance requirements that apply to a digital environment. This includes the following:

- Copyright
- Software licencing
- Spamming

We also look at the policies and procedures organisations can put in place to ensure compliance requirements are met, and that the associated risks can be avoided. This includes looking at what may be typically included in a Responsible Use Policy statement.

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Copyright

Copyright Act 1994

The Copyright Act 1994 sets out the copyright restrictions in New Zealand. Under the Act, any original work created, published and/or performed in New Zealand is automatically subject to copyright.



Copyright literally means 'the right to copy'. So, the person (or organisation) which holds the copyright for a particular item has the following exclusive rights:

- To copy, publish, distribute, perform, broadcast or adapt their work.
- To give up copyright on their work. Below are some examples for how a person might give up copyright to their work.



Creative Commons is where an author of some content (such as a graphic) licenses it to be used under certain conditions without the user needing to pay for it. These conditions usually include:

- **Attribution** - acknowledge the source of the content.
- **Share Alike** - indicate any changes you have made, and share the adapted material under the same conditions (licence) as the original.

Some Creative Commons licences only allow you to use the content:

- without modifications (ie without making changes)
- for non-commercial purposes (ie you are not allowed to use it for something that will make money).

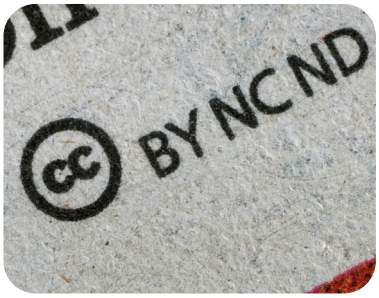
A list of the types of Creative Commons licences can be found at:

https://creativecommons.org/share-your-work/cclicenses/



Public Domain is a special type of Creative Commons licence which allows you to use images for **any purpose** without needing to acknowledge the source of the image.

There are two exceptions where the author is NOT the owner of the work:



- When an employee creates a work for their employer, then copyright belongs to the employer.
- When the work is commissioned by someone who pays for it. Then the copyright belongs to the person who commissioned the work.

For example, copyright of this learning guide belongs to Instant Education Solutions Ltd. It does not belong to the employee who actually wrote the guide.

There are some exceptions such as news reporting and education, where special permissions may be obtained.

While work subject to copyright usually displays the copyright symbol ©, the name of the copyright owner and the date, it is not necessary for copyright to apply.

Copyright (New Technologies) Amendment Act 2008

The purpose of this Act is to keep the Copyright Act relevant, as there have been large changes to technology since the Copyright Act came into force. An important change is the definition of copy to include digital copies.

Implications of these Acts for organisations

The consequences of breaking either Act can be severe: up to five years in jail or a fine of $150,000.

It is important to note that a director, manager and/or owner of a business can be held responsible if they:

- allow their employees to break copyright
- do not take steps to prevent their employees breaking copyright.



If you share something that you do not own the copyright to, you risk being imprisoned or fined. At a minimum, your ISP may disconnect your Internet account.

It is vital that organisations:

- provide employees with training and education related to copyright and ensuring that they do not intentionally or unintentionally break copyright requirements
- have in place policies and procedures related to copyright, and any requirements specific to the organisation
- have in place clear policies and procedures regarding the use and sharing of digital items in the workplace.

We will look at an example of what these policies and procedures may look like in the section on **Organisational compliance requirements**.

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Software licencing

Software programs fall under copyright. When you purchase software, you aren't paying for the software itself, you are paying for a licence to use it and you must keep to the conditions of the licence.

Some software is distributed for free. There are two main types of free software:

- **Open source software:** Software provided for use, modification, and redistribution. It can usually be downloaded from the internet at no cost. *Examples: Gimp – graphics manipulation software; Apache OpenOffice)*
- **Freeware:** Copyright still applies and is retained by the developer. *Examples: Skype; Adobe Acrobat Reader.*

Implications for organisations

Regardless of the specific type of licence under which you are using software, it is vital for organisations and individuals to stick to the licence terms for the software.

It is common for people to simply say they have read and/or accept the terms of a licence agreement, without actually reading them. This can lead to someone breaking the terms of the agreement unintentionally. However, they (and the organisation they work for) could still receive a severe penalty if they break any aspects of the licence agreement. There are three key ways in which licence agreements could be broken:

- **Organisation knowingly and deliberately** using pirated software and distributing it to staff.
- **Employee knowingly and deliberately** installing and using pirated software on the organisation's computer or network.
- **Using the software outside of the licence terms.** For example, this could include licence terms related to:
  - the number of devices on which the software may be installed
  - what the software may be used for, and what it may NOT be used for
  - how many people in an organisation can use the software
  - whether it can be used for commercial purposes (or only for personal use)
  - whether it can be used remotely (eg using a remote connection to the organisation's network).

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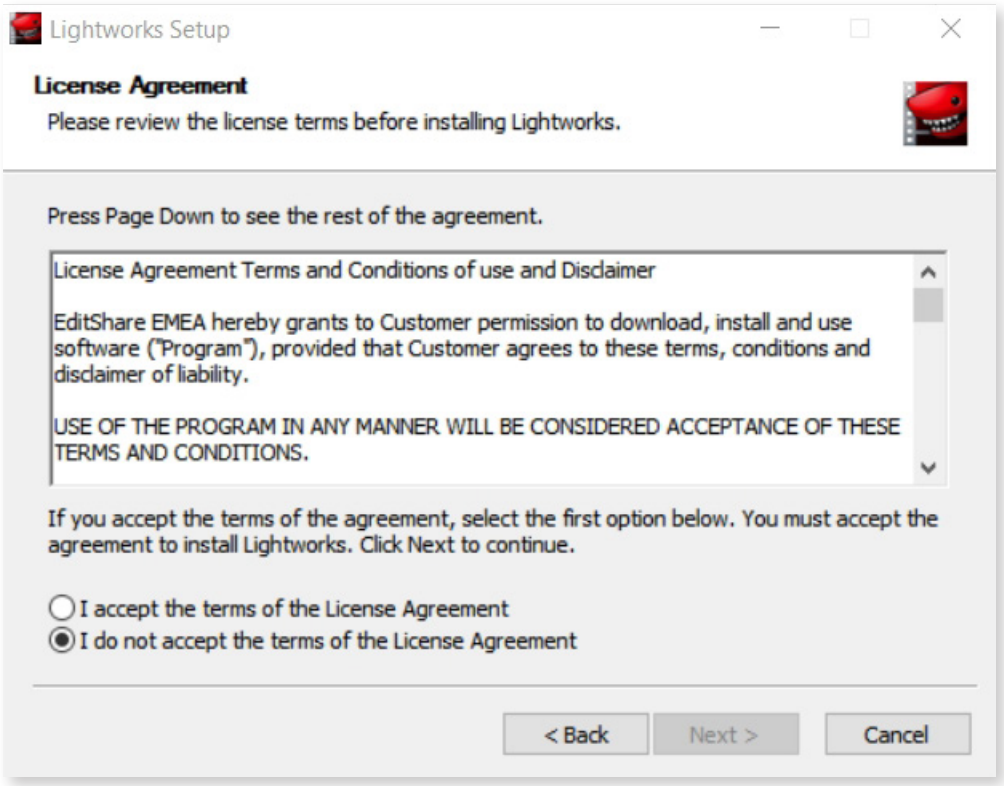
In addition to severe financial penalties for breaking licence agreements, there will also be a negative impact on the organisation's public image. This is because agencies who monitor software licence breaches make public the details of any settlements organisations have had to make because they have infringed a software licence agreement.

Steps an organisation can take:

- Stop users from installing software unless they have the correct permission.
- Include information concerning software in their IT policy.

Activity

Below is an example of a typical software licence agreement.



How many times have you simply selected 'I accept the terms of the Licence Agreement' without first reading through the agreement carefully?

Spamming

Spam is any unsolicited electronic message (ie a message you did not ask for). The **Unsolicited Electronic Messages Act 2007** prohibits spam being sent to, from, or within New Zealand. This includes e-mail, instant messaging, SMS and mobile phone messaging but NOT voice or fax messages.

The legal term for spam is 'unsolicited commercial electronic messages'.



For a message to qualify as spam it needs to meet all three of the following criteria.

- **Electronic** – includes email, Instant Messaging, SMS, multimedia message service, other mobile phone messaging.
- **Commercial** – includes messages which advertise goods, services, or land.
- **Unsolicited** – a message which is sent without the consent of the recipient.

Spam can be a major problem for organisations. Billions of spam messages are sent each year. This has several negative consequences.

NOTE: Although spam is often sent in bulk, a single message may be spam if it meets the criteria above.

Negative consequences of spam

- Slows down email delivery.
- Causes great irritation to the recipients.
- Reduces business productivity.
- Increases costs to organisations in the form of higher Internet access fees and the cost of installing and maintaining anti-spam software.
- Some spam messages:
  - try to trick people into revealing their credit card details, passwords or bank account information (ie phishing schemes – which you learnt about earlier in this guide)
  - spread computer viruses or contain false information.

Implications for organisations

All business must make sure that they are complying with the **Unsolicited Electronic Messages Act**. If they do not, there could be serious consequences for the organisation.

> **Example:**
> In February 2014, an Auckland-based marketing company (Image Marketing Group) was fined $120,000 for sending spam via email and text messages.

How organisations can comply with the Act

Below are some steps organisations can take to ensure they comply with the Act.

1. **Get consent from the recipient.** The person needs to give their permission for you to send them commercial messages.
2. **Clearly identify yourself.** The message needs to clearly indicate who sent it and what their contact details are.
3. **Include an unsubscribe facility.** The recipient needs to be given the option to unsubscribe at any time. In other words, they need to be able to say that they don't want commercial messages sent to them in the future.

   If someone asks to 'unsubscribe' this request must be actioned within five working days.

Organisational compliance requirements

Throughout this guide we have seen how important it is for organisations to have in place policies and procedures that help ensure they meet all legal and regulatory compliance requirements.

- **Policies** relate to what the organisation requires and the behaviour it expects from employees/learners.
- **Procedures** relate to what should be done and how it should be done. It could include procedures:
  - to minimise risks to the organisation
  - for what to do if something goes wrong (eg if there is a data breach, a health and safety problem arises, etc).

It is particularly helpful for organisations to have clear written organisational compliance requirements that are clearly communicated to all employees/learners.

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- Where necessary, the organisation needs to take all reasonable steps to ensure employees understand and know how to apply these organisational compliance requirements. This may involve, where necessary, additional education and training for employees.
- These organisational compliance requirements could be included in an employee's employment agreement, with the employee needing to sign to acknowledge that they have read and understood the requirements.

Responsible use policy

One specific type of policy that many organisations have in place is known as a **responsible use policy**. This may also be called a *fair* or *acceptable* use policy. This type of policy sets out the rules and guidelines for how a digital tool or system may be used within the organisation. Most organisations require people to sign the policy before they are given access to the organisations' digital tools and/or systems.

In a school, this might be one of the enrolment documents that needs to be signed when you enrol at a new school. In a business, you might have to sign a responsible use policy as part of your employment agreement.



Activity

Can you remember if you needed to sign a responsible use policy when you enrolled at your place of learning, and/or when you started as a new employee in a workplace?

Most responsible use policies include the following information:

- The purpose of the policy.
- What users may or may not do when using any digital tools or systems while at the organisation. This may include specific rules and guidelines related to the following.
    - Privacy and confidentiality
    - Health and safety

Continued on next page...

- **Copyright** and the **use of licenced software**
- **Security:** This usually deals with general guidelines for ensuring the security of the information systems within the organisation. It may include rules and guidelines related to regular software updates, especially for anti-malware software.
- **Data access and management:** This deals with methods and procedures to ensure that privacy, confidentiality, and security are maintained when accessing, using, storing, and transmitting data.
- **Internet use policy:** This may include the sites or Internet-based apps that may (or may not) be accessed within the organisation, using the organisation's hardware, software, or network.

MARKETING ACE

Marketing Ace – IT Systems Acceptable Use Policy

1. Purpose

This document outlines the authorised use of all Marketing Ace information technology and communication systems (ie IT systems). This includes the use of email, the Internet, voice, and mobile IT tools.

The policy applies to all:

- Marketing Ace employees, contractors, and agents (hereafter referred to as 'individuals')
- information, in whatever form, relating to Marketing Ace business activities worldwide, and to all information handled by Marketing Ace relating to other organisations or individuals with whom it deals
- IT and information communication facilities operated by Marketing Ace or on its behalf.

By signing this document, you agree to adhere to this policy at all times when accessing or using any IT systems tools (hardware, software, network). A breach of this policy is regarded as serious by Marketing Ace and could result in disciplinary action.

2. Access

Access to the Marketing Ace IT systems is controlled using User IDs, passwords and/or passcodes. All User IDs and passwords are uniquely assigned to named individuals and consequently, individuals are accountable for all actions on the Marketing Ace IT systems.

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Individuals must **not** do any of the following:

- Allow anyone else to use their user ID and password on any Marketing Ace IT system. This includes mobile devices such as smartphones and tablets.
- Leave their user accounts logged in at an unattended computer or other device.
- Use someone else's user ID and password to access Marketing Ace IT systems.
- Leave their password unprotected (for example writing it down).
- Attempt to access data that they are not authorised to access or use.
- Exceed the limits of their authorisation or specific business need to interrogate the system or data.
- Connect any non-Marketing Ace authorised device to the Marketing Ace network or IT systems.
- Store Marketing Ace data on any non-authorised Marketing Ace equipment or non-Marketing Ace device.
- Give or transfer Marketing Ace data or software to any person or organisation outside Marketing Ace without the authority of Marketing Ace.

Line managers must ensure that individuals are given clear direction on the extent and limits of their authority regarding IT systems and data.

3. Privacy and confidentiality

- Privacy refers to the sharing of personal information with other people. Marketing Ace takes privacy very seriously. We have strict expectations about the way you gather and deal with personal customer information during your work at Marketing Ace.
- Individuals must do the following:
    - Only gather personal information from customers as specified in the database.
    - Only share customer's and employee's personal information with people who are authorised to access this information. This includes other employees at Marketing Ace, permitted third parties as specified in the project list, and relevant government organisations.
    - Ensure customer information is stored securely at all times. For example:
        - Personal or confidential business information must be protected using security features provided, for example, secure print on printers.
        - When unattended, computers must be logged off/locked or protected with a screen locking mechanism controlled by a password.
        - Care must be taken to not leave confidential material on printers or photocopiers.
        - All business-related printed matter must be disposed of using confidential waste bins or shredders.

Continued on next page...

- Let management know if there has been a breach of customer personal information and the steps you are taking to resolve this breach.

- You must not do anything to interfere with or endanger the privacy of customers or other employees. This includes but is not limited to:
  - using digital tools with another employee's username and password
  - writing comments about a customer or employee in a way that breaches their privacy
  - printing or emailing information labelled 'confidential' without permission from management.

If you have any privacy concerns, please contact your team leader.

4. Working offsite

It is accepted that laptops and mobile devices will be taken offsite. The following controls must be applied when digital devices are taken offsite:

- Equipment and media taken offsite must not be left unattended in public places and not left in sight in a car.
- Laptops must be carried as hand luggage when travelling.
- Information should be protected against loss or compromise when working remotely (for example at home or in public places). Laptop encryption must be used.
- Particular care should be taken with the use of mobile devices such as laptops, mobile phones, smartphones and tablets. They must be protected at least by a password or a PIN and, where available, encryption.

5. Malware

The IT department has implemented centralised, automated malware detection and malware software updates within Marketing Ace.

All PCs have antivirus software installed to detect and remove any virus automatically.

Individuals must **not**:

- Remove or disable anti-malware software.
- Attempt to remove malware-infected files or clean up an infection, other than using approved Marketing Ace anti-malware software and procedures.

6. Copyright

The following highlights Marketing Ace's policy in relation to the ethical usage of Internet-based resources in relation to copyrigh:.

- Copyright refers to the right that creators of information have to their information. Creators of information can restrict who has access to this information.

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- Internet-based resources may be subject to copyright law. This means that the way you use, and have access to this information, is restricted.
- Information obtained from the Internet, or electronically on Marketing Ace's server, is restricted in the following ways:
    - Employees may not share any copyrighted information (including text, images, videos, diagrams or any other resources) with any other person or organisation. This means that the information may not be shared electronically or printed and given to others.
    - Employees may not save any copyrighted information (including text, images, videos, diagrams or any other resources) on any electronic server or hard drive. This includes Cloud servers such as Dropbox, OneDrive, or Google Drive.
    - Employees must not publish any unauthorised information copyrighted to Marketing Ace (including text, images, videos, diagrams or any other resource) without prior permission.
- In all reports and other documents created by employees at Marketing Ace, employees must use APA referencing. Information from third-party resources must be fully referenced.

7. Software

- Employees must use only software that is authorised by Marketing Ace on Marketing Ace computers.
- Authorised software must be used in accordance with the software supplier's licensing agreements.
- All software on Marketing Ace computers must be approved and installed by the Marketing Ace IT department.

Individuals must **not**:

- Store personal files such as music, video, photographs or games on Marketing Ace IT equipment.

8. Internet and email Conditions of Use

- Use of Marketing Ace Internet and email is intended for business use.
- Personal use is permitted where such use does not affect the individual's business performance, is not detrimental to Marketing Ace in any way, is not in breach of any term and condition of employment, and does not place the individual or Marketing Ace in breach of statutory or other legal obligations.
- All individuals are accountable for their actions on the Internet and email systems.

Continued on next page...

Individuals must **not** do any of the following:

- Use the Internet or email for the purposes of harassment or abuse.
- Use profanity, obscenities, or derogatory remarks in communications.
- Access, download, send or receive any data (including images) that Marketing Ace considers offensive in any way, including sexually explicit, discriminatory, defamatory, or libellous material.
- Use the Internet or email to make personal gains or conduct a personal business.
- Use the Internet or email to gamble.
- Use the email systems in a way that could affect its reliability or effectiveness, for example distributing chain letters or spam.
- Place any information on the Internet that relates to Marketing Ace, alter any information about it, or express any opinion about Marketing Ace, unless they are specifically authorised to do this.
- Send unprotected sensitive or confidential information externally.
- Forward Marketing Ace email to personal non-Marketing Ace email accounts (for example a personal Gmail account).
- Make official commitments through the Internet or email on behalf of Marketing Ace unless authorised to do so.
- Download copyrighted material such as music media (MP3) files, film and video files (not an exhaustive list) without appropriate approval.
- In any way infringe any copyright, database rights, trademarks or other intellectual property.
- Download any software from the Internet without prior approval of the IT Department.
- Connect Marketing Ace devices to the Internet using non-standard connections (eg using your smart phone to provide an Internet connection via a Personal Hotspot, connecting to an unsecured network, etc).

By signing this policy document, you indicate that you accept that any breaches of this policy or any New Zealand-based digital communications laws may result in your access to our server being removed. In situations where there has been a breach of New Zealand-based digital communications laws, you may be subject to legal prosecution.

To follow is an example of a responsible use policy. Read the policy and then answer the questions that follow.

Part of this policy document was adapted from the **Get Safe Online** Sample Acceptable Usage Policy. Retrieved from: www.getsafeonline.org.



Activity

1. The table below contains a list of items dealt with in the Marketing Ace Acceptable Use Policy document. Next to each item, indicate which section of the policy deals with the items. You only need to write the section number. In some cases, the item may be dealt with in more than one section.

   The first one has been done for you as an example.

Item	Section number
What the purpose of the policy is	1
Cybersafety (ie safe Internet and email use)	8
How to deal with viruses and other malicious software	5
How to securely store private and/or confidential information	3
Security	2, 3, 4
Software licensing	7
Spam	8
The access and use of copyright materials	6
The rules relating to using Marking Plus digital devices when you are not in the office	4
What should be done if there has been a privacy breach	3
What will happen if an employee does not adhere to the policy	8
Who and what the policy applies to	1

2. The table below contains a list of actions. According to the Marketing Plus Acceptable Use Policy, which of these things may an employee do?

   For each of the actions listed, put a tick (✓) next to those an employee may do, and a cross (×) next to those they may not.

Action	✓ or ×
Connect their personal mobile phone to the Marketing Plus network.	×
Download multimedia files if they first get permission from their manager to do so.	✓
Ensure there is access control (eg password or PIN protection) of all their mobile devices.	✓
Forward Marketing Plus emails to their Gmail account.	×

Continued on next page...

Gather personal information from customers as needed for the Marketing Plus database.	✓
If there has been a privacy breach, take steps to solve the breach, without telling management what has happened.	×
If they are somewhere they don't have access to broadband, use a personal hotspot on their mobile phone to get Internet access on their laptop.	×
Let another employee use their password.	×
Log off or use a screen lock when they are not using their computer for more than a few minutes.	✓
Print confidential information if they have been given permission by their manager to do so.	✓
Share Marketing Plus data with someone outside of the organisation, if their manager has said they can do so.	✓
Share Marketing Plus employees' personal information with relevant government organisations (eg the IRD).	✓
Store Marketing Plus data on their personal tablet.	×
Store their personal media files on a Marketing Plus mobile device.	×
To protect against malware, install their own anti-malware software and update it regularly.	×
Turn off the anti-malware software if it is blocking access to the Internet.	×
Use another employee's password to log on to the Marketing Plus network.	×
Use security features such as secure print when printing personal or confidential business information.	✓
Write their password on a sticky note to help them remember it.	×

3. What are some of the possible consequences to employees of **not** adhering to the Marketing Plus Acceptable Use Policy.

They might be subject to disciplinary action.

They may have their access to the Marketing Plus server removed.

If they also breach the law, they may be subject to legal punishment.

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Activity

1. Find a copy of your organisation's acceptable/responsible use policy. Keep in mind that your organisation:

- may have more than one policy document related to the acceptable/responsible use of digital tools
- may not have a **written** acceptable/responsible use policy.

2. Answer the following questions about your organisation's acceptable/responsible use policy.

   - Remember that if your organisation does not have a written policy, you need to get the information from your supervisor or other relevant person in authority (eg your teacher).
   - If you cannot find the answer to one of the questions below in your organisation's written policy, check with your supervisor or other person in authority.
   - Write the answers in your **own** words. Do not just copy everything directly from the policy. You can quote small parts from the policy documents, but you must make sure you clearly indicate the source of the quote.
   - If you do not understand something in the policy (eg a word that is used or an instruction that is given), make sure you take the time to find out what it means. Look up words you don't know in a dictionary (eg www.dictionary.com) and/or ask someone to help you if you don't understand something. It is important that you not only know *what* the policy is, but that you understand the policy and how to implement it.

a. What is your organisation's policy regarding **privacy** and **confidentiality**?

Continued on next page...

b. According to your organisation's policy regarding **security** of data and computer systems?

c. What is your organisation's policy regarding the use of **copyright** content?

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d. What is your organisation's policy regarding **software**?

e. What does your organisation's policy say regarding the **safe** use of digital tools?

Continued on next page...

f. What is your organisation's policy regarding **spamming**?

Teacher's Note:

This exercise is intended to help learners prepare for the assessment (especially Assessment Task 1, Part 2 and Assessment Task 2). The aim is that learners should not only know how to access their organisation's policies, but that they also understand these policies and how to implement them.

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LESSON 2:

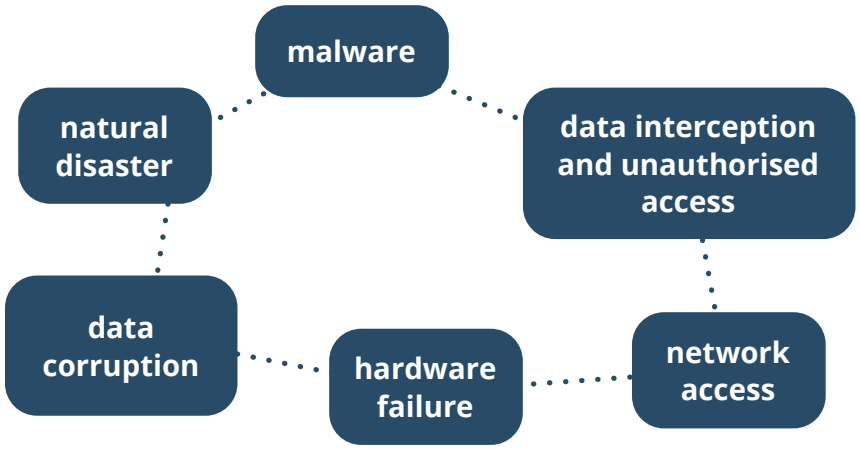
Addressing security and privacy risks and requirements

Learning Objectives

When you have finished this lesson, you will be able to explain and use a range of procedures and solutions to address security and privacy risks and requirements. This includes procedures and solutions related to:

- data interception and unauthorised access
- malware
- natural disasters
- data corruption
- hardware failure
- network access

In the previous lesson, we looked at the potential security and privacy risks related to digital information, and the implications of these for organisations. Now let's look at what can be done to address these risks. We will focus on how to address the following security and privacy risks.



To follow are some examples of procedures to address these security and privacy risks in terms of the impact this have on end users when using digital tools.

A note about good practice

For each of the procedures to address security and privacy risks discussed below, it is important to keep in mind good practice requirements. Good practice refers to selecting and using the appropriate feature or function to enable the safe and correct use of chosen digital tools.

Data interception and unauthorised access

People who are not authorised to access important or sensitive data are a big threat to the integrity of data. This ranges from the theft of confidential information, to the intentional damage to data, or the system.

We are going to look at some ways end users can help to prevent unauthorised access to devices and systems.

Passwords and other tools to restrict access

It is important that access to all devices is restricted using appropriate access restriction tools.

Many newer devices offer a range of options for securing digital devices. This could include using a PIN, pattern lock, fingerprint, eye scanner lock, voice recognition, or a password.

The type of security to be used will depend on what is available on the device and what the user finds most convenient. But what is most important is that the user ensures they use some form of security.

If a password is being used, it needs to be a unique, strong password.

- **Unique:** Don't use the same password for your device that you use for your accounts or applications. Using the same password makes it easier for a hacker to get into more places.
- **Strong:** Some guidelines for creating strong passwords are as follows:
    - Include a mix of numbers, punctuation, and upper and lower-case letters.
    - Preferably use a phrase (group of words), or a password that is at least eight characters long.
    - Avoid using repetition, dictionary words, sequences of letters or numbers, usernames, or personal information such as names or dates.

The way that you access the security settings on your device will depend on the type of device you are using.

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Organisations should have procedures and policies set in place that ensure anyone who works in the organisation, or links to the organisation's network, does the following:

- Understands how to create and use a secure password, or other method of access restriction, on ALL their devices.
- Changes their passwords etc regularly.
- Never shares their passwords etc with anyone else.
- Takes proper precautions to ensure no-one else gets their password without their knowledge or intent. For example, they need to know how to avoid phishing scams, other people seeing them enter their pin number or password, etc.



Activity

Teacher's Note:
Answers will be specific to the learner.

Answer the following questions in relation to the organisation that you work for (or your place of learning).

1. Are all users at your organisation required to have a username and password?

2. How often do passwords have to be changed?

3. Are there any rules about sharing passwords with others? If so, list these rules.

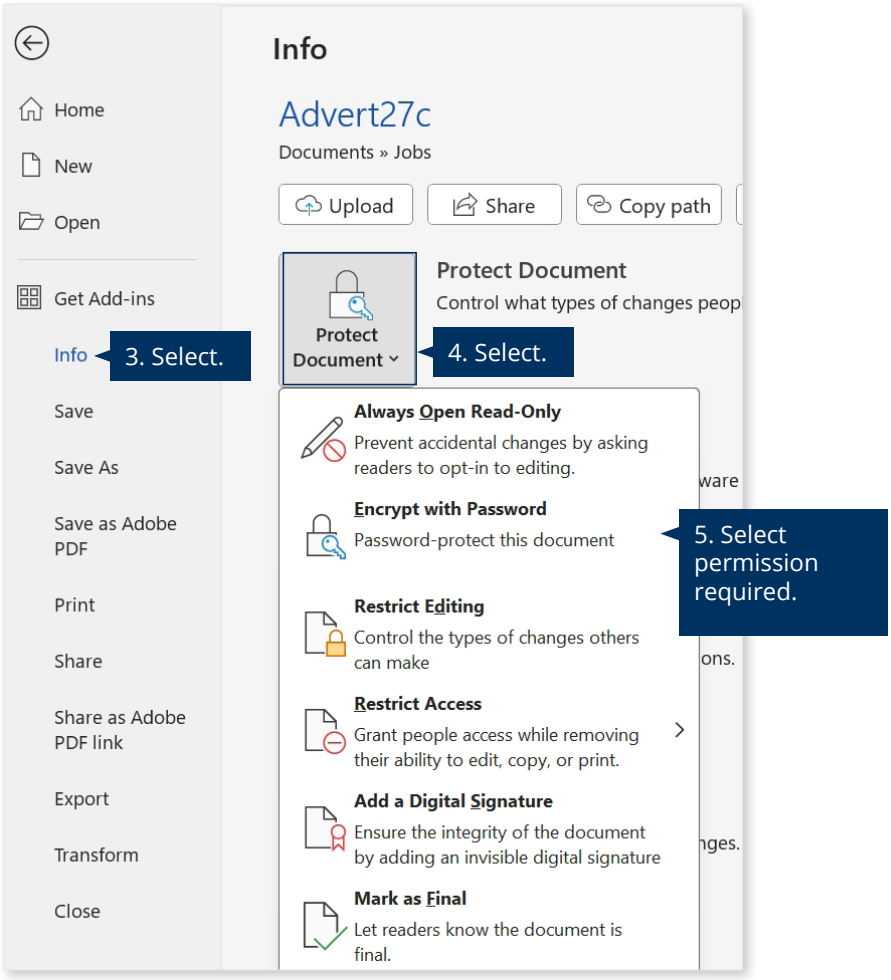
File access

The author of the file can apply permissions to the file. For example, you could do one of the following:

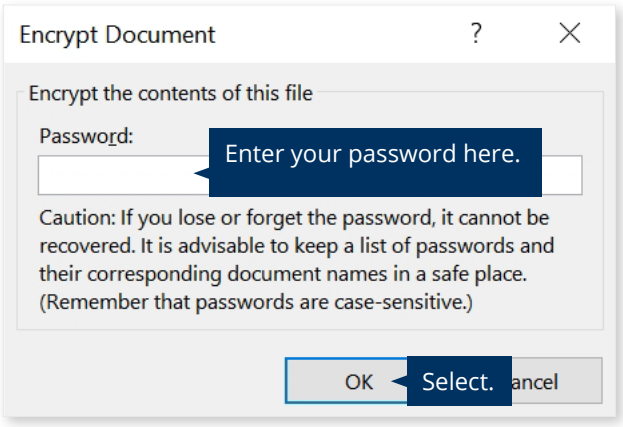
- Encrypt the file with a password so that only people who know the password can open the file.
- Give only certain people editing rights (this is known as **read only** access). In other words, anyone can read the file, but only someone with permission can make changes to the original file.
- Restrict the type of changes people can make to the document.

An example follows for how to apply permissions to a Word document. The process is similar for other types of files.

1. Open the file to which you want to apply permissions.
2. Select the **File** tab on the Ribbon.



If you selected 'Encrypt with Password' make sure you remember to enter your password.



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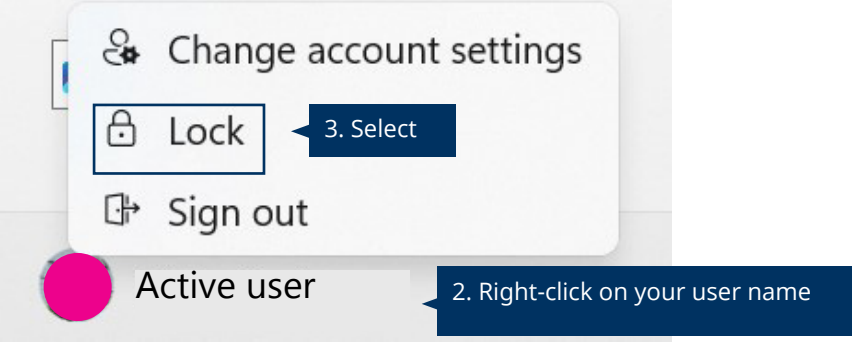
Computer security

Locking access to your computer

Some organisations may require end users to lock their computers whenever they are left unattended, even for a short time. When you 'lock' your computer, you prevent anyone else from using it. To 'unlock' your computer, you will need to enter your password.

To lock your computer:

1. Press the Windows key on your keyboard.



NOTE: You could use the following keyboard shortcut to log off:

Windows key + L.

Logging off

This is also known as signing out and refers to exiting out of your login altogether. Although the computer is still turned on, it can only be accessed by those who have an appropriate log-on username and password.

Your organisation may require all users to log-off the computer when they leave for morning tea, lunch break, at the end of the day, or to allow someone else, with a different login, to use it.

The difference between logging off and locking your computer is that:

- **log off/sign out** - you have exited out of the system and will need to use your user name and password to log back in. Someone else is also free to use their login details to access the computer.
- **locked** - you haven't exited the system and only need to enter your password.

To log off your computer, you follow the same process as for locking your computer, except that you select 'Sign out' from the dropdown menu (Step 3 above).

Shutting down

Shutting down a computer refers to turning it off. Some organisations require all computers to be shut down at the end of the day.



Activity

In the boxes below, write when you might lock your computer and when you would log off.

Lock a computer	*Going to the bathroom* *Going to lunch* *Collecting printing from the printer* *Going to a meeting*
Logging off	*At the end of the day* *If I am not using the computer anymore* *If someone else would like to use the computer and has a different profile.*

All three of these security measures will decrease the chances of someone gaining unauthorised access to your computer, documents, or taking advantage of your private information such as bank account details.

Physical security

The fewer people who have access to the place where important and sensitive data is stored, the less likely that someone will use the data in a way that they shouldn't. Making sure that only authorised people have physical access to data, will help protect its integrity.

Some organisations coach their staff to challenge unfamiliar people accessing buildings or departments. Others have locked rooms with special access requirements.

There may also be organisational requirements that people physically lock away their computers or other devices at the end of the day and when they are away. For organisations that supply mobile devices (eg smartphones, tablets, etc) to their employees or members, there may be specific requirements about how and where these devices need to be stored when the user is offsite (ie away from the office, etc).

It is also a good idea to ensure computer and other device screens do not face towards public areas. For example, in a reception area, the computer screens should be turned so that a visitor to reception is not able to see what is on the screen.

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Staff training

Staff training is a key requirement related to the security of data (**Principle 5** of the Privacy Act). As has already been mentioned, staff training is important to ensure staff know how to set and use passwords and other access restrictions effectively.



Organisations should also have in place training that deals with other aspects of security and privacy. This should include training that ensures end users understand:

- why it is important to keep data secure and accurate, and to take all steps to prevent unauthorised access
- who may access and use specific data, and who the data may be shared with
- that they should not access data that they are not authorised to access
- who they should inform if they become aware of a data breach.

Policies and procedures related to sharing digital information

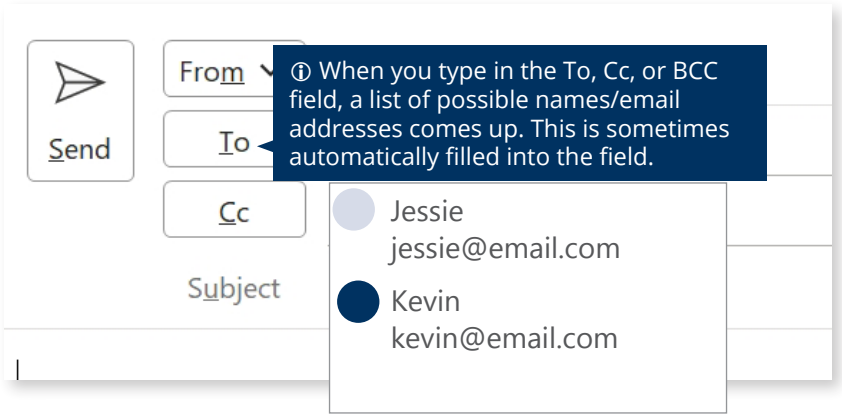
Organisations should have clear policies in place regarding what information may be shared digitally and how this should be done. Below are some examples.

Information that may be shared digitally

Users should be aware of the type of information that may and may not be sent by email, SMS, or services such as Skype, etc.

Ensuring the correct email address is entered

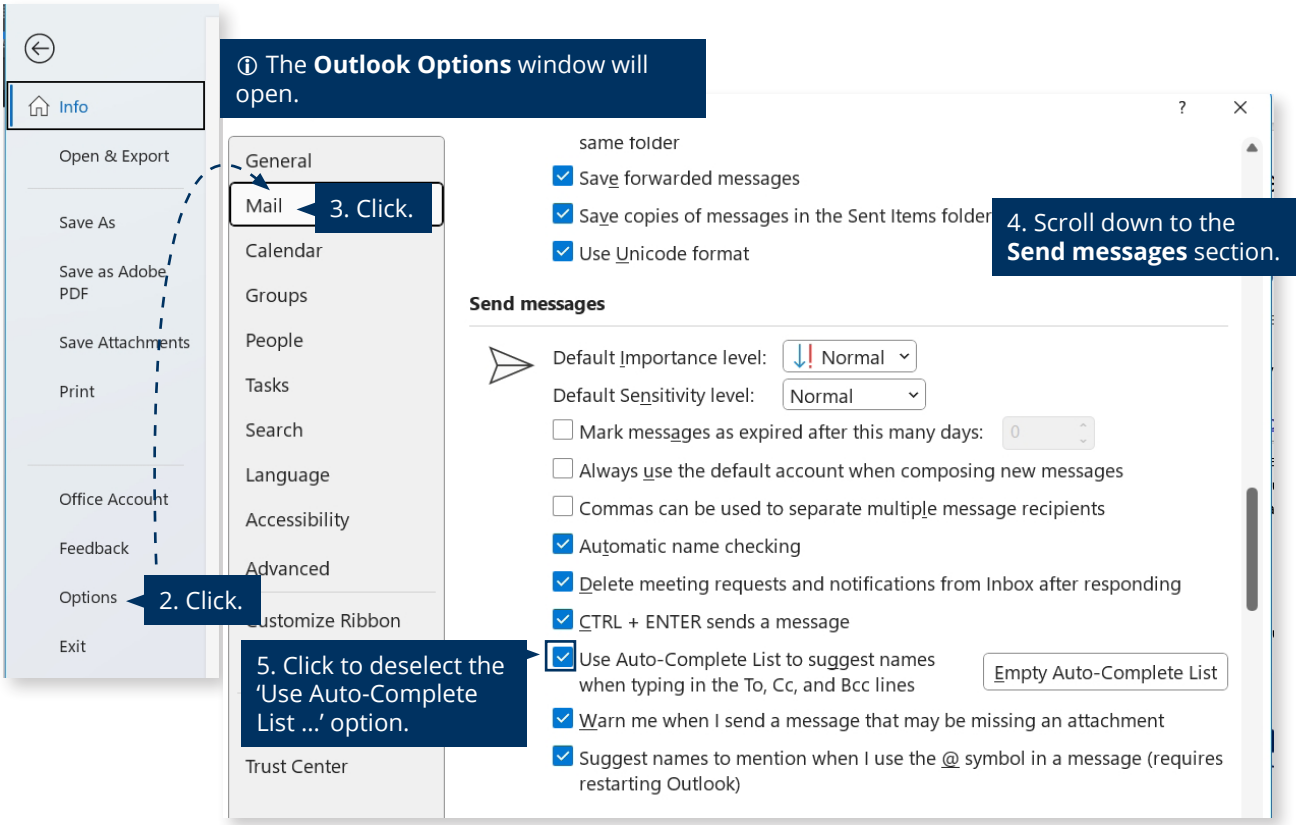
Great care should be taken when entering email addresses to ensure the correct email address is used. Some email software has an auto-complete email name list function. This works by suggesting a name when you start typing in the To:, Cc:, or Bcc field. Look at the example below:



If you do not carefully check the results of the autocomplete, your email could end up going to the wrong person (eg it could go to Sam Smith, instead of Sam Woods).

For this reason, some organisations have a policy that autocomplete should be deactivated. Below is an example of how to do this in Microsoft Office Outlook.

1. Click on the **File** tab on the Ribbon.



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Forwarded emails

You need to take great care when forwarding an email, to ensure that no unintended information is sent in the message.

Example: Private information forwarded in error

In the email below:

- Andrew works in the finance department.
- Sam is a new employee.
- Karen is arranging the staff Christmas party.

From: Andrew Mitchell
Sent: Wednesday, 19 July
To: Karen Louann
Subject: FW: Contact details

Hi Karen
See below – Sam is interested in helping out regarding the Christmas Party.
Thanks
Andrew

From: Sam Smith
Sent: Wednesday, 19 July
To: Andrew Mitchell
Subject: RE: Contact details

Hi Andrew

Below are my contact details:

Physical Address: 12 Marsden Lane
Lunaville
Mob: 056 9387 495
Home phone: As for mobile
Email address: sam@waccess.com

While I am talking to you: I heard that a staff Christmas party is currently been arranged. I just wanted to find out more about when the party is planned for and if I can help to arrange it.

Thanks
Sam

From: Andrew Mitchell
Sent: Wednesday, 19 July
To: Sam Smith
Subject: Contact details

Hi Sam

Could you please provide me with your contact details (physical address, home and mobile phone number, email address), so that I can update your employee records.

Thanks
Andrew
(Finance Department)

As you can see from the email, Andrew only intended to tell Karen that Sam is eager to help with the Christmas Party. However, because he forwarded the entire email chain, Karen now has access to Sam's personal information, which is a privacy breach.

Using disclaimers and confidentiality notices

Disclaimers are a way of limiting an organisation's responsibility in relation to possible risks when sharing data.

Many organisations include a disclaimer in all outgoing email messages (ie email messages sent outside of the organisation).

Below is an example of an email disclaimer addressing the risks linked to corporate confidentiality and privacy (also known as a **confidentiality notice**).

Confidentiality notice

This email and any files transmitted with it are confidential and intended solely for the use of the individual or entity to whom they are addressed. If you are not the named addressee you should not disseminate, distribute or copy this e-mail. Please notify the sender immediately by e-mail if you have received this e-mail by mistake and delete this e-mail from your system. You are hereby notified that any use, dissemination, distribution and/or reproduction of this message and/or any attachments by unintended recipients is unauthorized and may be unlawful.

Using confidentiality agreements

Many organisations require employees or anyone else given access to sensitive organisation information to sign a confidentiality agreement, or a confidentiality clause in their employment contract. This is a written, signed agreement between the organisation and the employee that defines confidential information and in which the employee agrees that they will not share this information with anyone else.

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Malware

Anti-malware apps

Organisations should have procedures in place related to the use of anti-malware apps, to protect against malware attacks. This could include:

- All devices should be running a good anti-malware program that covers both viruses and other malware.
- Anti-malware software should be set to scan devices regularly, as well as any new emails and files that are installed.
- The software should be regularly updated.

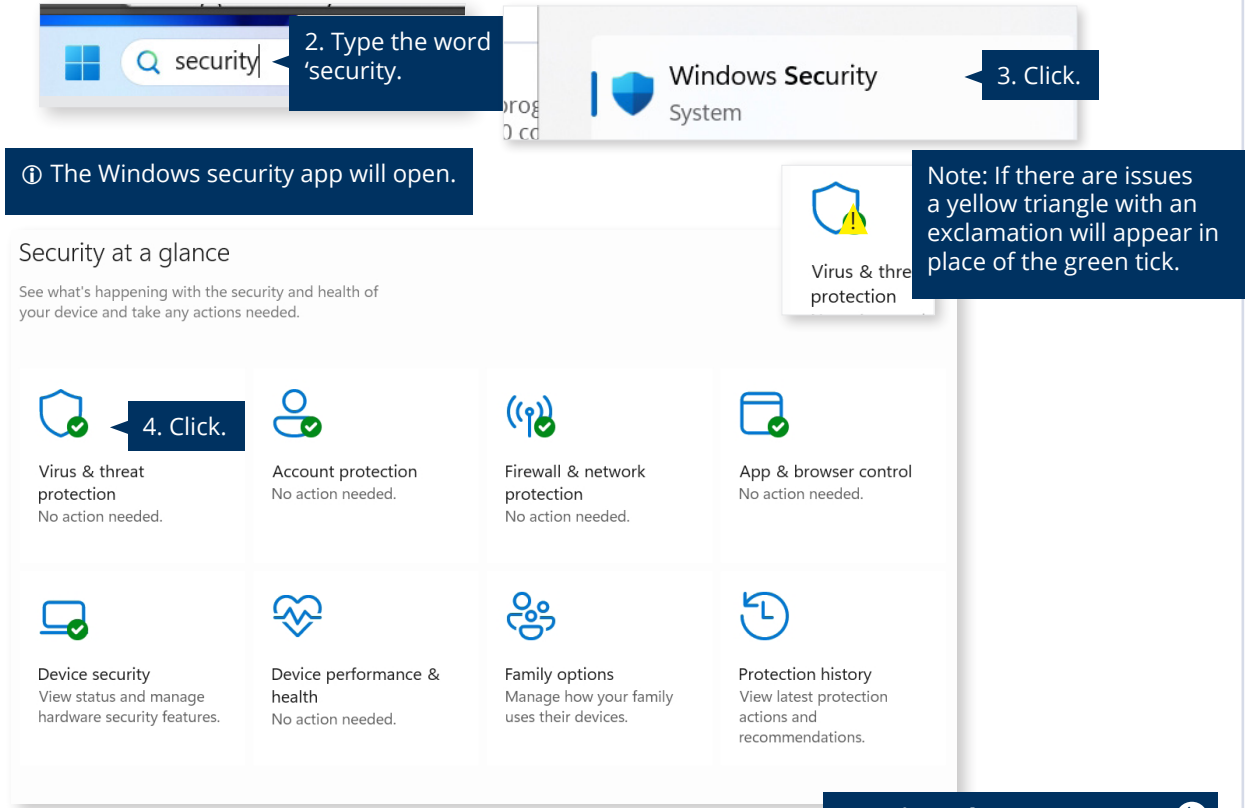
Windows Security

Windows 11 comes with an inbuilt anti-malware program. If you are using a Windows 11 computer, and have no other anti-malware software, Windows Security should automatically be turned on.

The instructions below are for how to check Windows Security is running, and then use it to scan for malware.

Checking and using Windows Security

1. Press the ⊞ key on your keyboard.

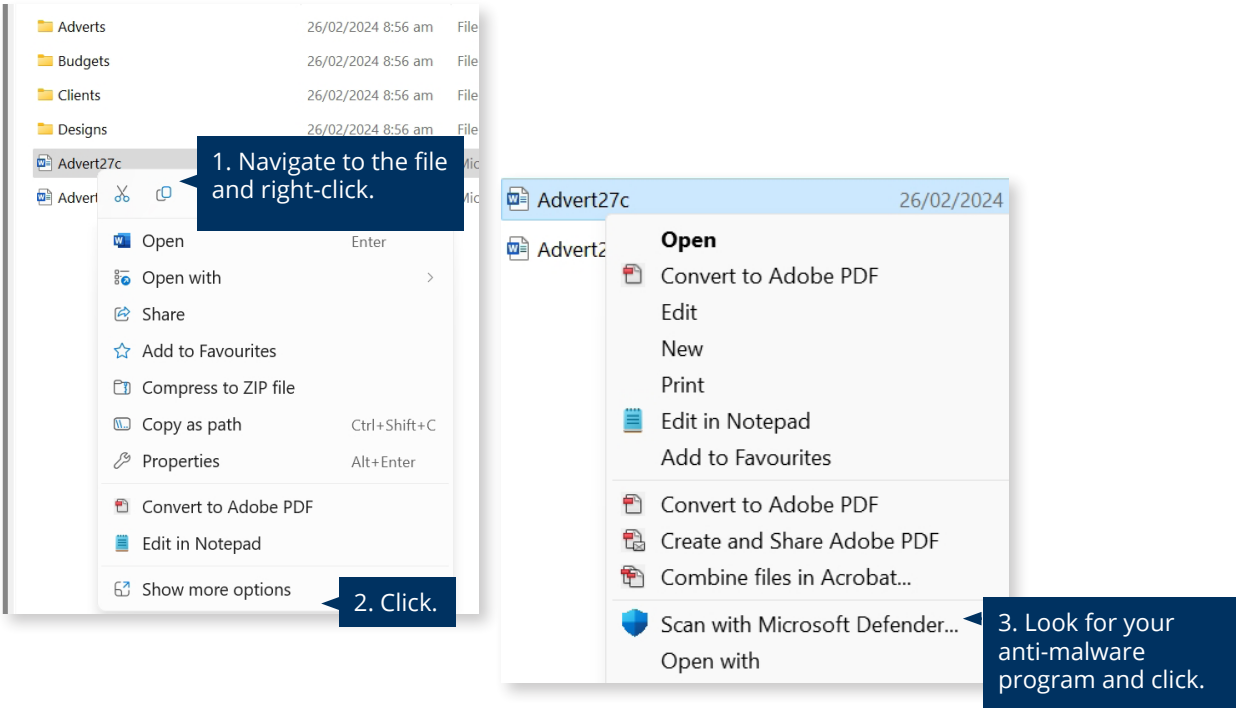


Continued on next page...



Scanning an individual file

You can scan individual files in File Explorer:



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- Use a firewall. Windows 11 comes with an inbuilt firewall which helps block unauthorised access to your computer. Refer to the section on **Network Access**, later in this lesson, for more information on firewalls.
- Portable devices such as flash drives should be scanned (using your anti-malware app), before a file is opened or copied to your PC.
- Do not open suspicious emails or their attachments.
- Do not use public computers when you need to enter personal information, eg social media, Internet banking.
- Be VERY careful what you download from the Internet. Only download and install software from trusted sources.

Symptoms of a malware infection include:

- Sending emails the user did not write.
- Crashing and rebooting frequently.
- Menus and dialogue boxes look strange.
- Files are missing or damaged.
- Running slower than usual.

If the person thinks their device has a virus, they should:

- disconnect it from any network it is on
- run their anti-malware software to do a full malware scan
- immediately inform the relevant person in the organisation that their device may be infected.

If there has been a data breach (data is lost, stolen, damaged, etc) because of a malware attack, refer to the previous section for what to do to manage a data breach.



The NZ Police have useful information and links on their website (http://www.police.govt.nz) about how to protect yourself and your organisation from malware attacks. Click on the 'Advice' tab, and then select 'Email and internet safety'. Alternatively, search for 'Email and internet safety' using the search facility on the website.

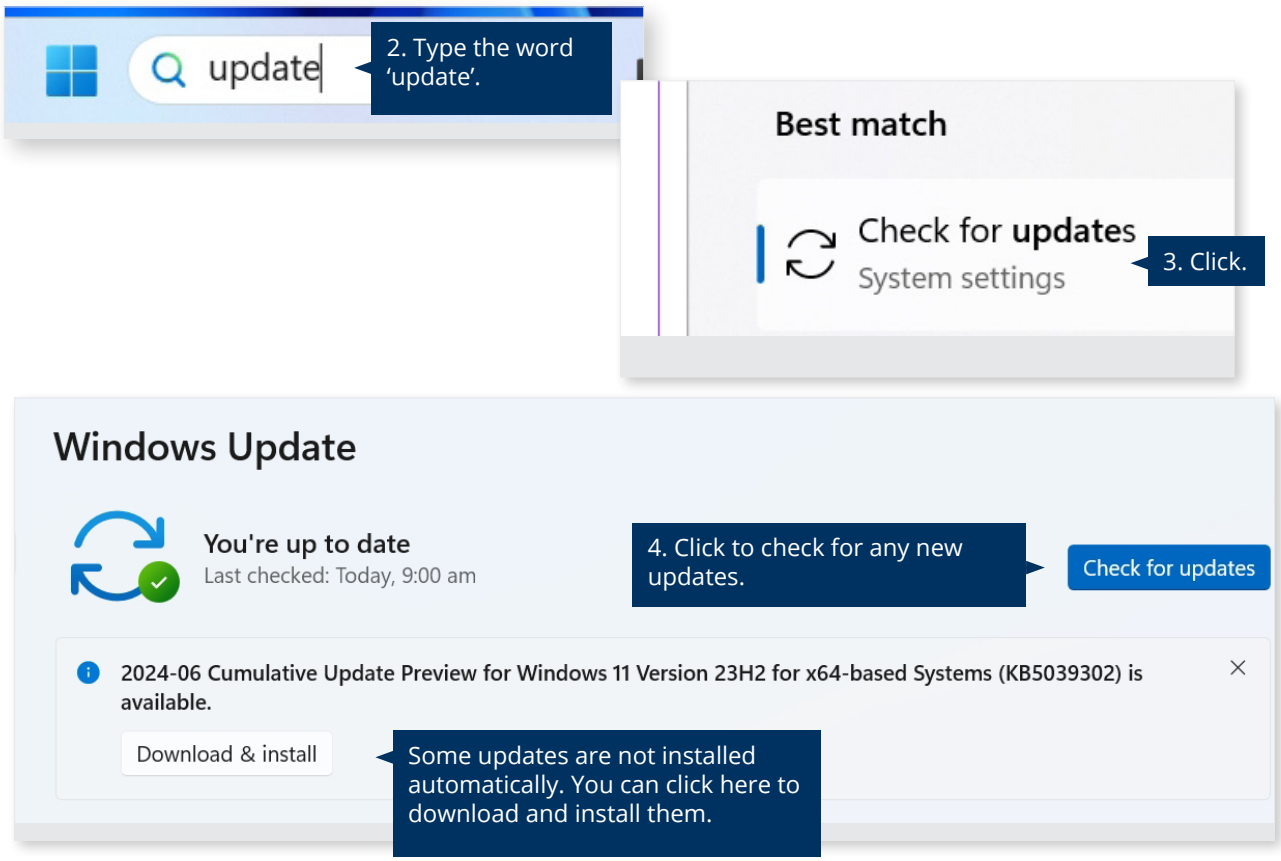
Regular operating system updates

In the same way that regular updates of your anti-malware software can help prevent malware attacks, it is also important to regularly update your operating system (eg Windows, Android, iOS, etc) to ensure the best protection against malware.

This is because all software systems have flaws or vulnerabilities in them. These are ways that hackers and viruses can use to inflitrate your device. To prevent this happening, software manufacturers regularly make updates available which help to fix these security vulnerabilities. These are usually in the form of operating system (OS) patches known as bug fixes or security patches.

On most devices you will be given a notification of when a new update is available. It is important that when you get such a notification, you update your OS as soon as possible. Usually the notification itself will contain the information you need to perform the update. Otherwise, you can do a manual update via your device settings menu. An example is shown below for updating the Windows OS.

1. Press the key on your keyboard.



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Browser settings

Browsers such as Google Chrome, Firefox, and Microsoft Edge are used to access content on the World Wide Web. As such, they are a potential source of security vulnerability for your devices. Some organisations have rules regarding which browsers may be used. This is because some browsers are considered to be more secure than others.

Regardless of what browser you use, you need to set your browser's security and privacy settings to help reduce the risk of a data breach or malware attack.



Activity

Security and privacy settings for different browsers are accessed in different ways. Do a Google search to find out how to change the security and privacy settings for the browser(s) you use.

For example, if you use the Google Chrome browser, you could use the following search terms to find out how to change the security and privacy settings.

google chrome privacy security settings

In general, you should ensure that you adjust the settings to do the following (where possible):

- Enable automatic updates for the browser.
- Block pop-ups, plug-ins, and phishing sites.
- NOT store passwords.
- Disable third-party cookies.
- Any other specific browser security and privacy settings your organisation requires (note these below).

_____

_____

_____

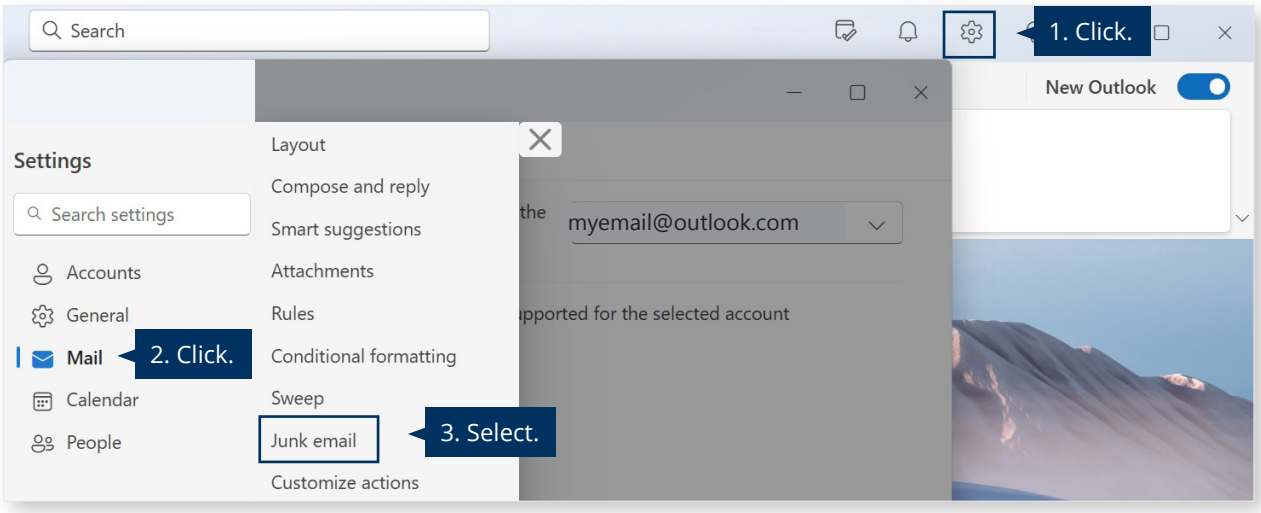
You can also include any of the above keywords in your Google search. For example, if you are using Google Chrome:

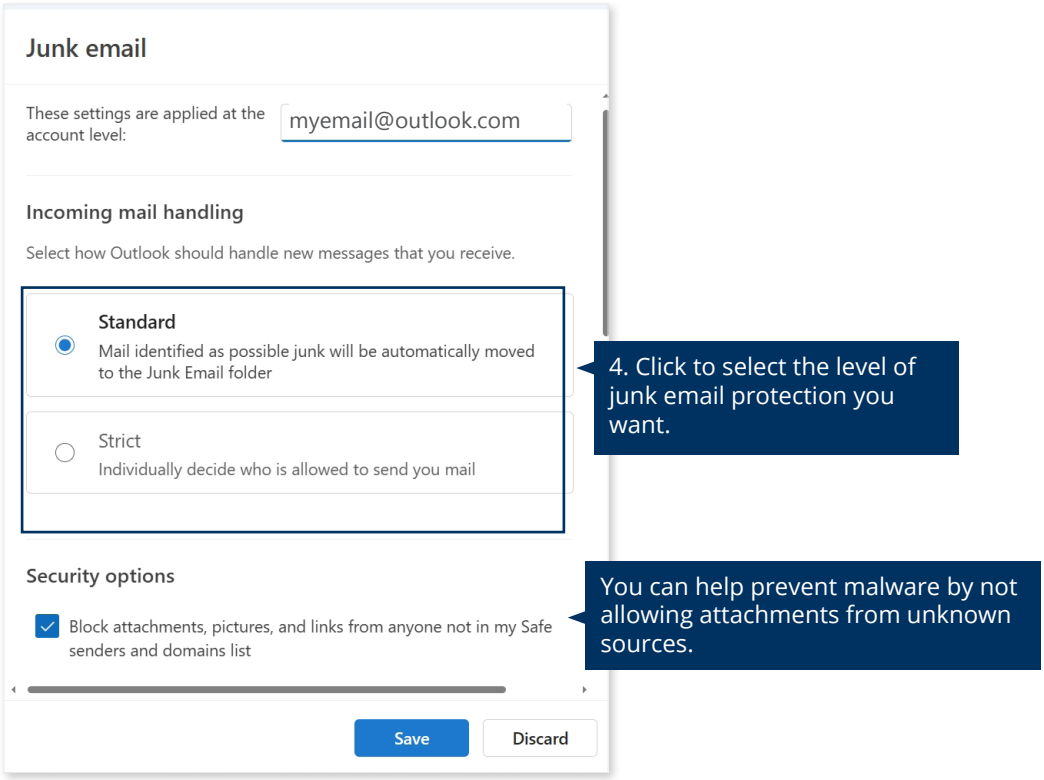
google chrome block pop-ups

Email settings

We have already looked at how disabling autocomplete of email addresses, and using disclaimers and confidentiality notices can help address privacy risks associated with sending emails. Setting appropriate Junk email settings can also help to guard against spam and phishing scams. This in turn will help to address the risk of a malware attack via an email (eg a phishing email).

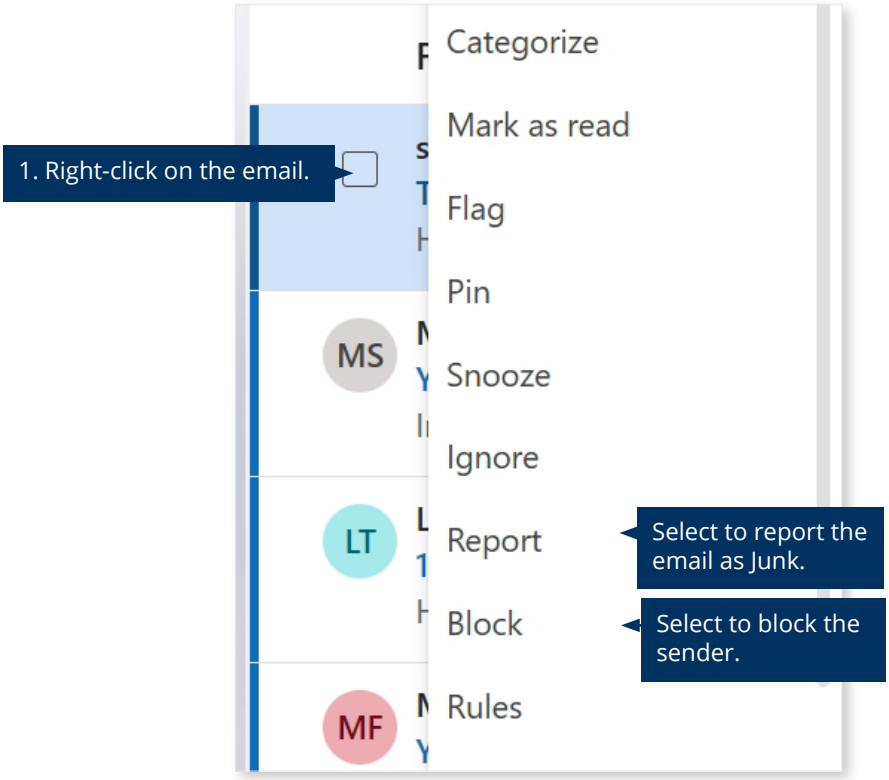
Below is an example of how to adjust Junk email settings, using Microsoft Office Outlook.





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If you receive an email that you suspect is junk email (or spam), you can block it by entering their address via the tab shown on the previous page. Otherwise, you can do the following:



If you block the sender then you will no longer receive emails from them.

Emails that look legitimate

Sometimes you will receive an email from someone whose email address you recognise, but it still has some of the common signs of being a phishing scam or spam. This may indicate that the person's email address has been hacked. In a case like this:

- do not click on any links in the email
- immediately contact the person (preferably by phone or text message) to find out if they did send the message
- if they did not send the message, ask them to immediately inform everyone in their contacts list/address book that their account has been hacked and not to open the spam/phishing email.

Natural disasters

Some organisations have disaster recovery plans that guide them through natural disasters so they can get back to 'business as usual' as soon as possible. They may also store their backups offsite (ie not in the organisation's buildings) so they do not lose all their data if there is a natural disaster.

When offsite storage of backup is not possible, well-designed buildings, as well as alarm and sprinkler systems can help to limit any damage.

Most organisations, including workplaces and schools, will have their own management plans for backing-up and restoring data. These may outline different types of backup, and disaster recovery plans.

These plans may also include the storage of backup data by an online backup service. An online backup service allows important data to be backed up over the Internet and securely stored on a server in a professional data centre offsite.

An organisation could also back up its data to the Cloud. You can back up your own data using Cloud services such as Google Drive, OneDrive and Dropbox.

Teacher's Note: Steps for how to perform local and external (Cloud-based) backups is covered in depth in the **Instant Education Solutions** resource for US29772v2 (Manage files and folders using digital devices).



Activity

What does your workplace or school do to back up its data? Does it back up onsite, offsite or to the cloud?

Teacher's Note: Answers will be specific to the learner.

What does your school/workplace require you to do in terms of saving your work and backing up your data? If there are no written policies regarding saving and backing up your work, check with your manager or teacher.

Teacher's Note: It is important that the learner is aware of their organisation's backup plan and policies. For example, it may be against policy to copy data onto remote devices.

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Data corruption

Many of the measures that can be taken to address unauthorised access and malware risks, also apply for dealing with the risk of data corruption. Important protections that reduce the risk of data corruption include:

- appropriate training so that data entry is accurate
- read only files, so that only authorised people can make changes to files
- backing up data regularly
- security software to prevent data corruption by viruses, etc
- using UPS or surge protectors (see below for more information).

UPS and surge protectors

Interruptions to the mains power supply to your computer could cause it to crash or corrupt its drives, resulting in the loss or corruption of your data.

In addition to regularly saving and backing up your work, two pieces of electrical equipment could help you if there is trouble with the power supply. These are:

- an Uninterrupted Power Supply (UPS)
- surge protectors.

UPS

A UPS (Uninterrupted Power Supply) gives your computer a continuous power supply. If the power is disrupted, it gives the user a few minutes time to save their files and shut down their computer.



Surge protector

Surge protectors (or surge suppressors) provide some protection against sudden changes in the power supply that could cause your computer to crash. An electrical surge is a temporary, very short (less than a second) increase in the electrical supply voltage (or current, or both). This could be the result of damage to a power transformer due to an accident or lightning strike.

The surge protector works by blocking or by shorting to ground any unwanted voltages above a safe threshold. Unfortunately, it won't be able to protect your computer against direct lightning strikes to the computer itself.

Hardware failure

To guard against the risk of hardware failure, organisations should have policies and procedures in place to ensure:

- all devices are kept in good condition and regularly checked for potential problems (eg dust built up in the computer box)
- users don't spill liquids on their devices
- users have adequate physical protection for mobile devices (eg smartphone and tablet cases to protect against damage from the device being dropped; water-resistant cases if devices need to be used outdoors; etc)
- users immediately report any strange sounds or behaviour by the device
- users only move a desktop computer if they know how to do so safely and correctly
- all files are regularly backed up to an external storage location (either a removal storage drive or offsite storage) - in case there is a serious hardware failure and the computer is permanently damaged.

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Network access

In the section on malware, we already looked at several procedures that also apply for reducing risks associated with network access. One of the key ways to protect computer systems from threats from the Internet, is the use of a firewall.

A firewall works to protect your computer against threats from the Internet. This includes restricting or preventing the spread of computer viruses and Trojans.

A firewall controls the incoming and outgoing data based on a particular set of rules. It establishes a 'wall' between a trusted, secure internal network – your workplace or schools computer system – and another network that cannot be trusted (ie the Internet).

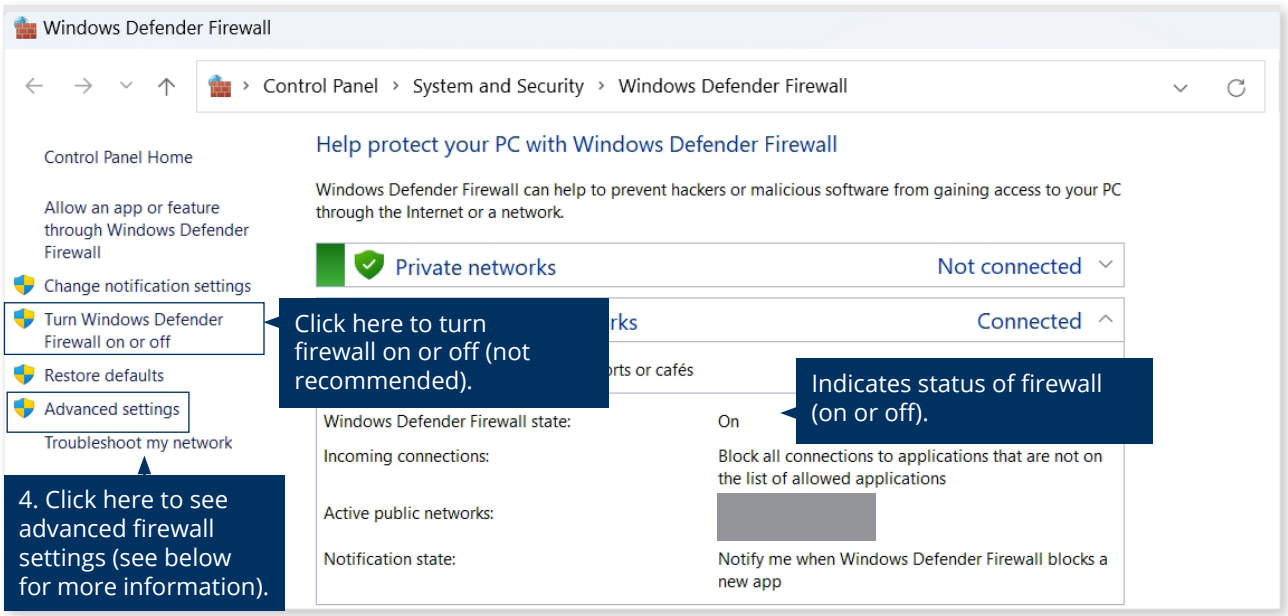


Below are instructions for how to access the inbuilt Windows 11 firewall settings.

1. Press the ⊞ key on your keyboard.

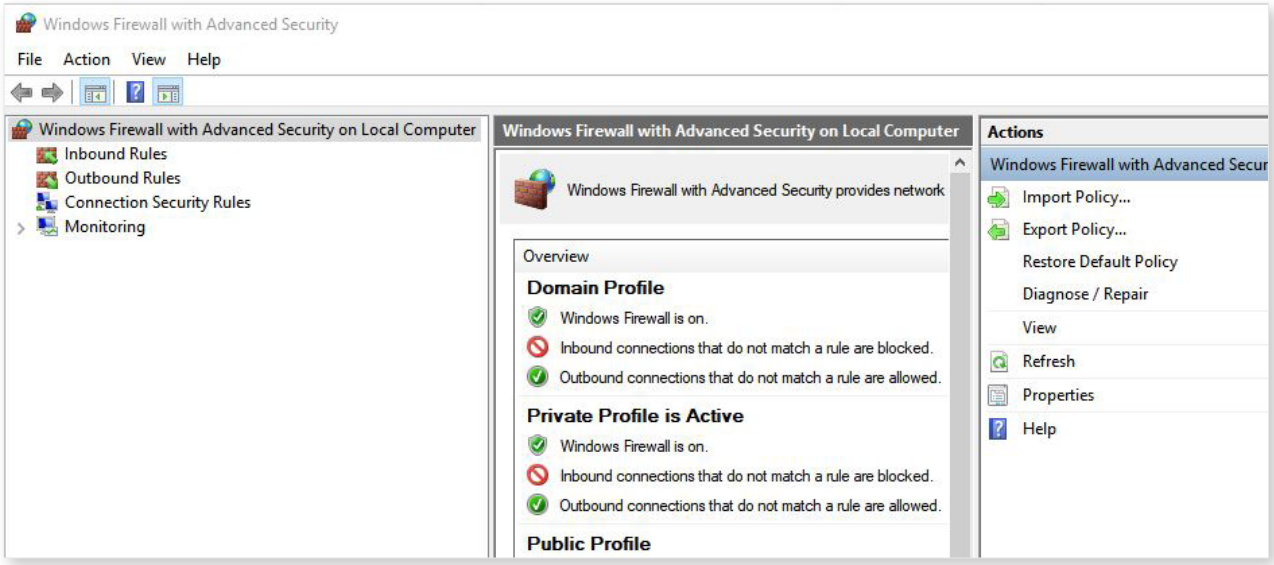


The Windows Firewall settings (in Control Panel) will open (see below).



When you click on 'Advanced settings', a window like the one below will open. Here you can see more detailed information about the firewall settings (centre panel). You can also adjust and set inbound, outbound, and connection security rules (left panel).

NOTE: In most contexts, the default Windows firewall settings are appropriate. You should not make changes to these firewall settings unless you have been specifically instructed to do so by your organisation.



LESSON 3:

Health and safety

Learning Objectives

When you have finished this lesson, you will be able to identify and describe the following as it relates to the use of digital tools:

- Legal and regulatory compliance requirements related to health and safety.
- Potential health risks and safety issues linked to the use of digital tools.

In the previous lesson, we looked at privacy and security in a digital environment.

In this lesson, we look at the compliance requirements and risks related to health and safety when working in a digital environment.

A key piece of legislation (ie law) that applies to health and safety in the workplace is **The Health and Safety at Work Act 2015 (HSWA)**.

In this lesson, we will look at the compliance requirements related to health and safety, and the implications for organisations. In the lesson that follows, we will also look at the procedures and solutions that you can take to ensure your own health and safety, and that of others.

The Health and Safety at Work Act 2015 (HSWA)

The HSWA sets out the requirements, duties, and rights related to workplace health and safety. The aim of the Act is to help ensure the health and safety of workers and others (eg customers, clients, visitors, etc), in a workplace.

It outlines specific duties that both organisations and workers have to ensure a safe and healthy working environment. (See the section on **Implications for organisational processes** for more details about these duties.)

Guidelines for Using Computers (ACC)

The ACC guide: *Guidelines for Using Computers – Preventing and managing discomfort, pain and injury*, provides specific guidelines related to health and safety in a digital environment. It provides good practice guidelines for workers and organisations regarding the use of computers and other digital tools. We will look at some of these guidelines in greater detail in the next lesson.

Risks to organisations

There can be very serious implications for organisations and workers who do not take the necessary steps to ensure health and safety in a workplace. Fines can be as high as **$300,000** for workers and in the **millions** for organisations.

Organisations that do not take all reasonable steps to protect the health and safety in their workplace face more than just legal penalties. They also face the risk of:

- high employee absenteeism (because of illness, injury, etc)
- high employee turnover – people do not like to work in an unsafe and unhealthy environment
- customer/client dissatisfaction – people do not like to visit or buy things in an unsafe and unhealthy place
- loss of reputation – for not taking the necessary care to ensure the health and safety of workers, customers, clients and visitors.

NOTE: For further information on the Act see the **Health and Safety at Work Act 2015 – Special Guide**. This guide provides detailed information and explanations about the HSWA. It can be accessed online at: www.worksafe.govt.nz or by scanning this QR Code.

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Implications for organisational processes

Both the organisation and employees have responsibilities. In the context of a digital workplace, the following responsibilities apply:

Organisations

- Provide employees with equipment that is suitable for purpose eg adjustable chairs.
- Ensure the environment (lighting, ventilation, temperature, noise) are at acceptable levels.
- A high stess work load can cause fatigue and stress related illnesses. Employers should ensure that the workload is managable for their employees.
- Ensure employees are adequately trained in safe use of their equipment
- Monitor risks in the environment on an ongoing basis. This may include:
  - ergonomic risks such as strain, fatigue, or pain from repetitive motions, poor lighting, etc
  - physical risks such as noise, heat, exposure to chemicals (eg from a printer).

Employees

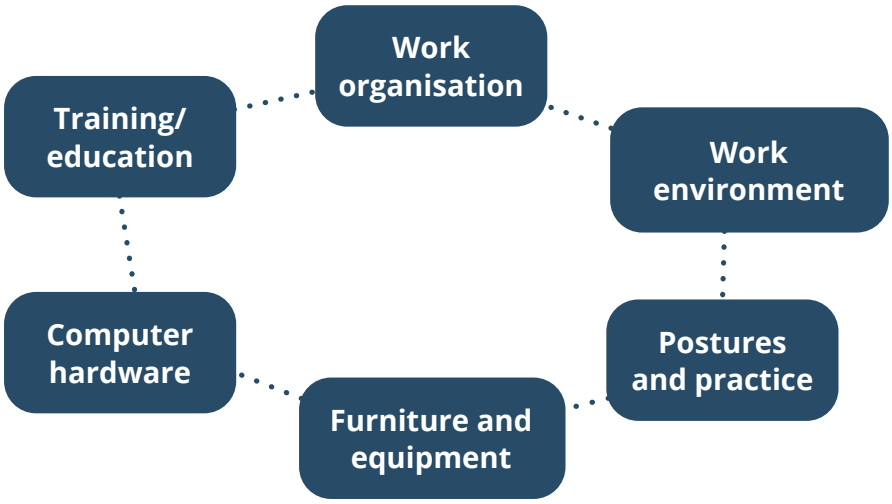
- Take reasonable steps to care for their own health and safety.
- Take reasonable steps to avoid affecting the health and safety of others
- Follow any reasonable health and safety instructions, policies and procedures their organisation has in place, and which they have been informed about.

Let's look at some specific examples of health and safety risks associate with working in a digital environment.

Safety issues related to working in a digital environment

There are countless health risks and safety issues that could be present in an organisation. The type of health and safety risks will depend on the type of service or product delivery the organisation is involved in. For example, the risk to a worker in the building industry is very different to that for an office worker (eg secretary, receptionist). In this section, we will focus on the health risks and safety issues most common to people working or studying in a digital environment. **NOTE**: Wherever you see reference to a 'worker' in the following sections, this can also refer to someone studying in a place of learning.

The ACC *Guidelines for Using Computers* identifies the following six possible hazards (or safety issues) related to working in a digital environment.



Work organisation

- The way work is structured and managed can have an impact on both the psychological and physical health of workers. Below are some examples.

Psychological

- Work schedules that are too demanding or work that is mentally or emotionally challenging, can cause stress for workers.
- Tasks that are too repetitive, monotonous and boring could lead to workers feeling frustrated or undervalued (which can have a negative impact on their psychological health).

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Physical

- Work schedules that require a worker to spend too many hours at their desk working on a computer, for example, can also result in physical health issues such as discomfort, pain, injury and fatigue.
- Tasks that require repetitive motions (eg some types of data entry), could also result in physical discomfort, pain, injury and fatigue.

Work environment

The lighting, noise levels, temperature and other aspects of the working environment can lead to physical discomfort or fatigue for workers. We will look at aspects of the work environment and how it can impact on health and safety later in this lesson.

Work postures and practices

The way that a person positions their body when working (eg how they sit at a desk), can have a significant impact on their physical health. Poor postures and practices can result in physical discomfort, pain, and injury, as well as fatigue. We will look at this in more detail in the section on **Health issues related to working in a digital environment**.

Selection and arrangement of furniture and equipment

The type of furniture and equipment used, as well as the way it is arranged, can also impact on physical health.

Selection

For example, a desk that is too small for the person using it, can cause them to sit in an awkward position while working. This can lead to discomfort and strain.

Arrangement

Ergonomics refers to the arrangement and use of a computer workstation (eg desk, chair, etc) in a way that helps to prevent and manage discomfort, pain and injury.

- **Good ergonomic** arrangement and use of a computer workstation helps to reduce the risk of strain or injury for the user.
- **Poor ergonomic** arrangement and use of a computer workstation can lead to different physical health problems. We will look at this in more detail in the section on **Health issues related to working in a digital environment**.

Selection and arrangement of digital device components

Digital device components (eg mouse, keyboard, etc) can also have an impact on the physical health of workers.

Selection

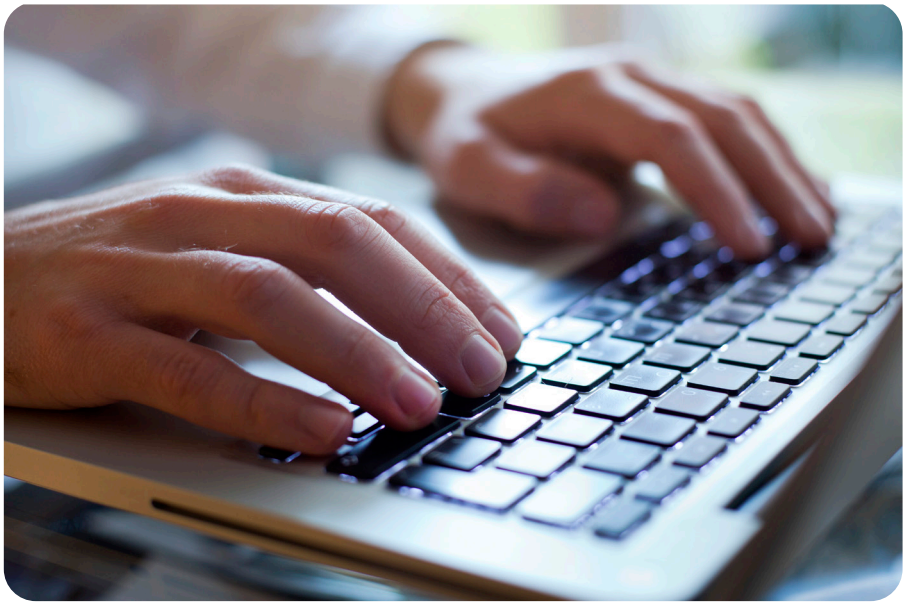
Workers who work long hours entering data may need a special, ergonomically designed keyboard and/or mouse, a screen that reduces visual discomfort and fatigue, etc.

Arrangement

In the same way that the arrangement of the computer workstation can impact on the physical health of workers, so can the arrangement of computer hardware such as the screen, keyboard, mouse, etc. See the section on **Health issues related to working in a digital environmen**t for more detail.

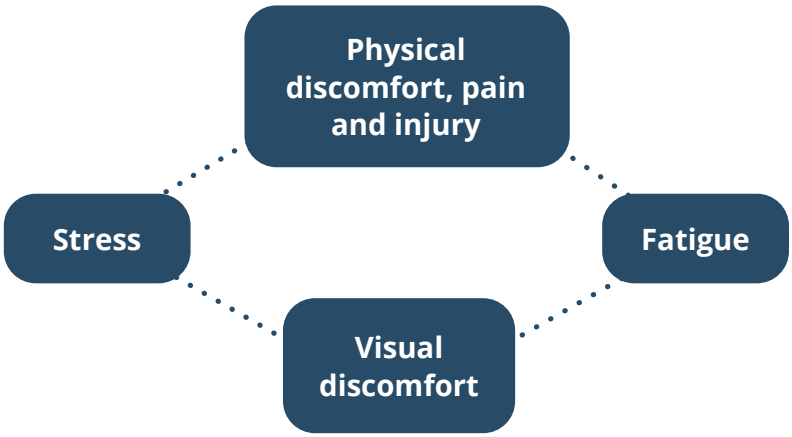
Training and education of workers

Training and educating people who work in a digital environment is very important. This can help them to recognise the potential hazards and understand how to reduce these risks. Workers who are not properly trained are at greater risk of computer-related health problems.



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Health risks related to working in a digital environment



It is important to note that these health problems are often linked and one type of health problem can lead to another type(s) of health problem(s).

Let's look at each of these problems and their common symptoms.

Physical discomfort, pain and injury

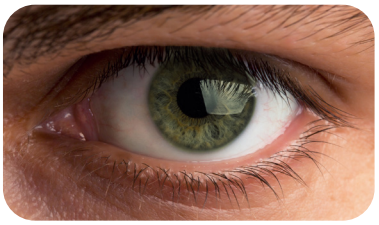
Poor posture caused by equipment design or workstation layout; or stress causing muscles to tense up, can cause a range of physical symptoms:



- pain
- muscle discomfort
- fatigue
- stiffness
- burning or tingling
- weakness or numbness

In the next lesson you will look at how your workstation, workflow and equipment can be adjusted to reduce the risk of injury.

Visual discomfort



Just like the muscles in other parts of the body, the muscles in the eyes can become tired and strained. People who regularly use computers for work or other purposes, often end up staring at a computer screen for long periods of time. This can cause a visual discomfort for the user.

Common symptoms of visual discomfort include the following.

- Sore, red, dry or watery eyes.
- Eyes feeling 'gritty' (like they have something in them) or 'heavy'.
- Blurred vision.
- Headaches (or even migraines).

As with other types of physical health problems, visual discomfort can have a number of other causes than computer work. However, visual discomfort caused by computer work often increases during the work day and then gets a lot better after work.

Activity

Have you ever experienced physical or visual discomfort or pain when using a computer?

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Stress

Stress is a common health problem experienced by most people at some time in their lives. Stress can be caused or made worse when people are asked to do something in their job that does not match their knowledge, resources, or abilities. It can also be caused when the user feels that they are not able to control or cope with a situation or they don't have the support they need.

Below are some common symptoms of stress.

- Physical aches and pains.
- Finding it difficult to:
  - relax
  - concentrate
  - sleep
  - think logically
  - make decisions.
- Feeling:
  - increasingly upset and unhappy
  - irritable
  - tired
  - depressed
  - less committed to work.
- Not enjoying work as much anymore.



Activity

How do you know when you are feeling stressed? What symptoms do you have?

Fatigue



Fatigue is another word for 'tired' or 'exhausted'. There are three main forms of fatigue - physical, mental, and emotional.

Physical fatigue in computer users is often the result of over-using the postural, arm, and other muscles involved in sitting and working at a computer.

Mental fatigue is when your brain starts to get tired because you are concentrating on a job for a very long time without taking a break from that task.

Emotional fatigue can result when you have to do a mentally challenging job and at the same time there are things going on which are emotionally stressful. For example, trying to work when you are worrying about relationship problems you are having.

Common symptoms of fatigue may include the following:

- Not being able to respond to a situation (as well as before) because you are 'tired out' from doing too much (ie over-activity). This over-activity could be physical, mental or emotional.
- Being constantly tired.
- Feeling weak.
- Lacking energy.

In the next lesson, we will look at the procedures and solutions that can be used to address potential health and safety risks and issues in a digital environment.

Activity

Have you ever experienced any of the symptoms of stress and fatigue when using a computer? What happened? Why?

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LESSON 4:

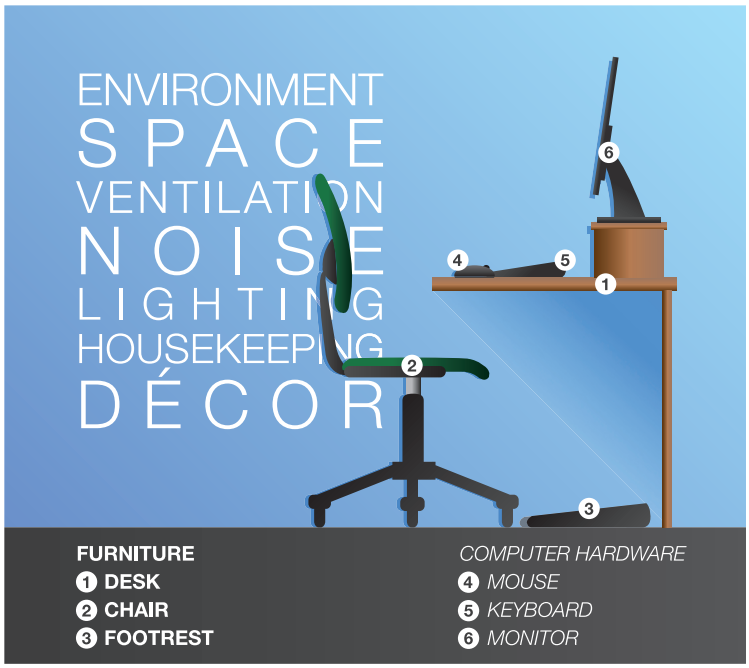
Safe work practices

Learning Objectives

When you have finished this lesson, you will be able to explain and use a range of procedures and solutions to address health and safety risks and requirements. This includes procedures and solutions related to the following:

- A safe and healthy working environment.
- Safe and healthy postures and work practices in a digital environment.
- The ergonomic arrangement and use of a computer workstation.
- The ergonomic use of digital tools.

In the previous lesson, we looked at risks and compliance requirements related to health and safety in a digital environment. In this lesson, we will look at procedures and solutions that can help to address these risks and requirements. The focus will be on procedures and solutions specifically related to ensuring your personal health and wellbeing and preventing harm, when working in a digital environment. This includes the working environment, posture and work practices, computer workstation, and use of digital tools.



Working environment

There are several factors that can impact on a safe working environment. Some of these are summarised below.

Space

- People need to have enough space to be able to do their work safely and comfortably.
- Amount of space needed will depend on the individual, their type of work, and hardware being used.
- Personal space is important so people don't feel 'crowded'.
- There needs to be enough space for a person to have a range of comfortable working postures.
- Users need to be able to get to/from their computers easily, without anything blocking their way.

Housekeeping

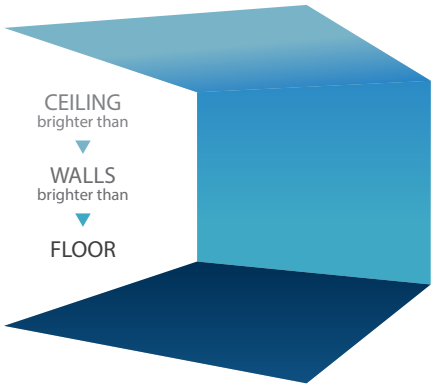
- Floors need to be kept tidy.
- Access ways (ie to and from workstation, etc) need to be kept clear.
- Power and other cables must not be a tripping hazard.

Noise

- Noise can make it hard for people to communicate and/or concentrate.
- Noise levels should be as low as possible.
- Noise levels should be monitored to be kept at an acceptable level.

Décor

- Décor affects the lighting in a room and so should be planned along with the lighting.
- Generally, ceilings should be brighter than walls; walls should be brighter than floors.
- Avoid sharp contrasts in brightness (eg a white ceiling and very dark wall).



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Atmospheric conditions

- The room should be well-ventilated. This is to help ensure that any air-borne pollution is removed.
- The room should be heated to a comfortable temperature.
- Most people find the following temperatures comfortable:
    - In Summer - between 23°C and 25°C.
    - In Winter - between 20°C and 24°C.

Lighting

Below are some common issues related to lighting which can cause problems for computer users.

- **Poor lighting** (ie not light enough, or too bright): This can lead to eyestrain and headaches.
- **Glare** or **shadows**. Light shining on your screen, eg from a lamp or a window can cause glare.
- **Contrast** (difference) in the brightness of the screen and the surrounding environment, this can cause eye strain. Having a window in front of you can cause this - using blinds may help.

One way to avoid these problems is to position the workstation correctly in relation to light sources. Below are some general guidelines.

- If possible, put workstations at a right angle to the light source.
- Try to position the desk so that shadows and glare are not caused by overhead or other lighting.

Later in this section you will see how lighting can also affect the position of the screen.

Printer position

There are several potential hazards associated with printers such as noise and fumes.

- If possible put the printer in a different room or somewhere with partitions.
- Make sure there is good ventilation.

Furniture & workstation layout

General guidelines

Workstation furniture and equipment needs to be arranged in a way that allows the user to:

- work comfortably, in a neutral position with good posture.
- move easily and change their body position regularly.
- do several different types of tasks - eg looking at things on the screen, typing on the keyboard, using the mouse, writing, and reading.

To follow are some more specific guidelines related to common workstation furniture and how they should be arranged.



Activity

1. What furniture do you most often use with a computer (eg chair, desk, bed!)?

2. What other types of workstation furniture have you seen or used before?

3. When using a computer, have you ever felt pain or discomfort because of the furniture, or the way it was arranged? What happened? Why do you think it happened?

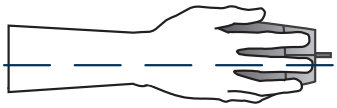
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Posture

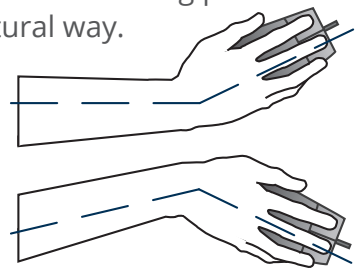
Posture is the way in which you position your body.

Having a good posture when using a computer can help reduce the chances of you suffering from strain, pain, or injury.

A **neutral body position** helps to reduce stress and strain on your muscles and bones. It is a comfortable working posture where your joints are lined up in a natural way.

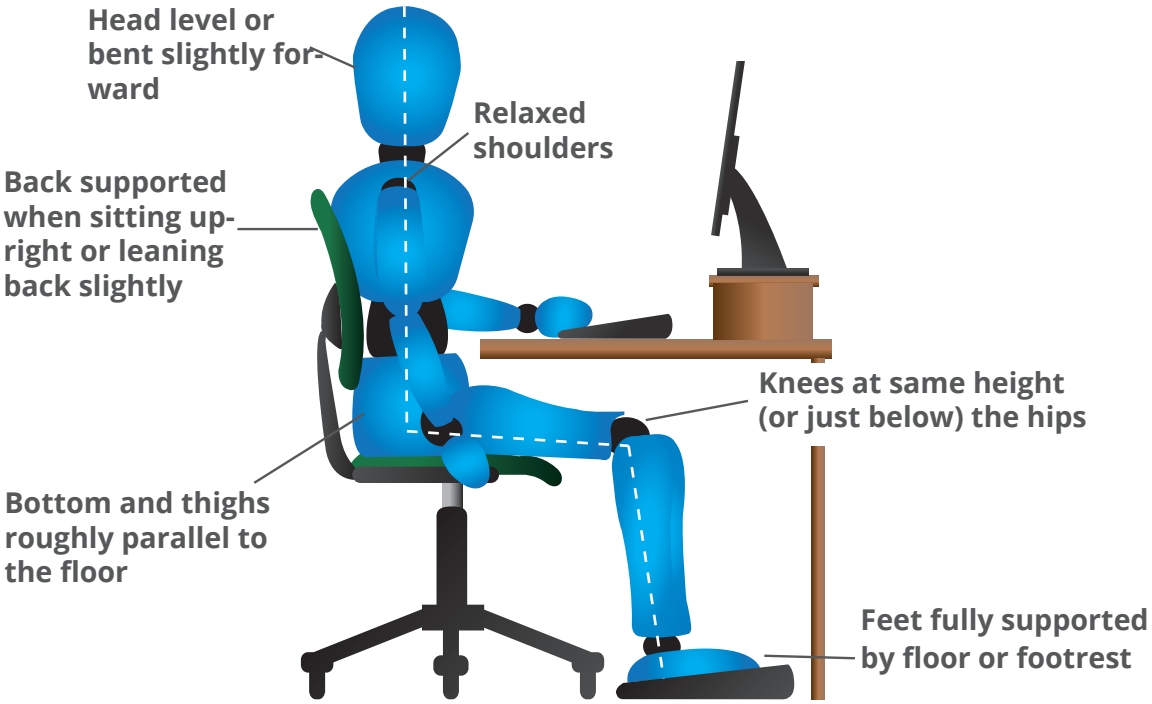


1. Fingers, wrist and forearm lined up in a natural, comfortable way



2. Wrist at an awkward angle to the forearm. This can cause stress and strain on the wrist and arm.

The diagram below gives some more specific guidelines for an ergonomic body posture. It is important to have an adjustable chair, and if possible an adjustable desk, in order to achieve a comfortable posture.



Desks

The desk height, width, depth, and thickness need to allow for the user to:

- comfortably get their legs under the desk
- move around
- work without feeling cramped
- sit in several different positions to comfortably use the computer equipment.

Chairs

The chair should be adjustable so that different users (of different shapes and sizes) can:

- have a comfortable posture in the chair and use as little muscle effort as possible to keep that position
- use the computer comfortably and without strain
- change their posture with as little muscle effort as possible.

The chair should have a backrest that:

- provides good support to the user's lower back
- fits the natural curve of a user's spine
- doesn't limit the user's upper body movement.

The chair should have a seat that:

- allows the user to have a comfortable leg and back posture
- is wider than the user's hips
- is wide enough to allow the user to move easily

The chair should have cushioning that is:

- soft enough to be comfortable
- firm enough to provide good support for several different body positions.

Footrests can help users to get a comfortable foot and leg position. For example, if a desk or chair can't be adjusted, then a footrest can be helpful to make sure the user can have a proper, neutral posture.



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Digital device components

To follow are some guidelines for the positioning and use of common digital device components. However, keep in mind that these are guidelines only – not rules or regulations. You might need to do things slightly differently, depending on your circumstances. **NOTE**: The focus is on desktop computers, but similar principles may also apply for other digital tools.

As you read each set of guidelines, check your own workstation, and the way you work at it. Make sure you have arranged, and are using, your digital tools according to ergonomic principles.



Activity

1. What types of digital devices do you use most often (eg desktop computer, laptop, tablet, mobile phone)?

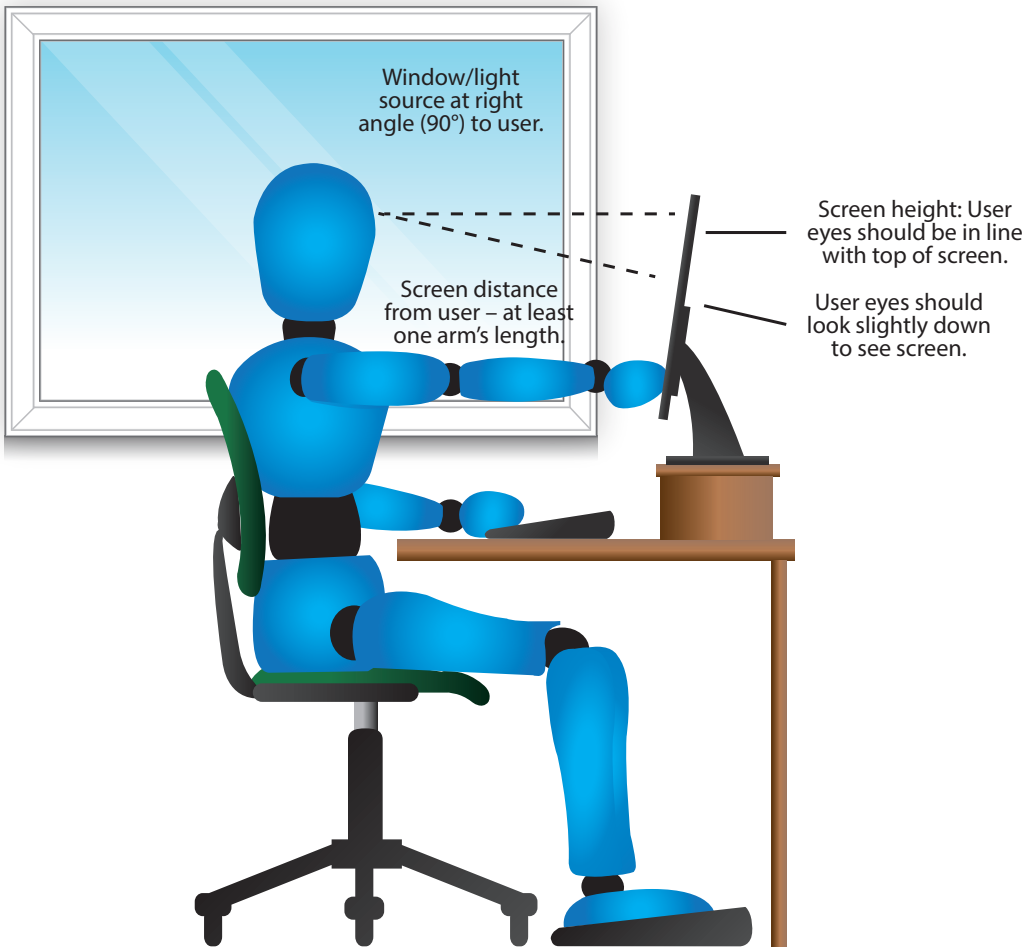
2. What types of components do you use with this computer (eg mouse, keyboard)?

3. Can you think of a time when you used computer hardware and felt pain or discomfort, or were injured? What happened? Why do you think this happened?

The screen

Your screen should be placed at a distance and height that does not strain your eyes. Ideally, it should be at arms length with your line of sight at the top of your screen.

Any light source such as a window should be from the side, not behind or in front of the screen to avoid glare on your eyes or on the screen.



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Activity

Look at the diagram on the previous page and answer the questions below. Use your own experience of working on a computer to help you.

1. What type of physical strain or injury could you get if the computer screen is too far away or too close to you?

Eyestrain because it is not easy to read text if it is too close or too far away. Neck and/or backstrain if you have to lean in to see the screen.

2. What could happen if the screen is higher or lower than your eye level?

Because you have to tilt your head up (or down) to see the screen, this could result in neckstrain and/or backstrain. It can also be a strain on your eyes to have to look up (or down) at the screen.

3. The guidelines show that (ideally) the user (and their screen) should be at right angles (90°) to a light source (eg a window).

   a) What potential problem (strain/injury) could result if the window was directly in front of you?

The light shining directly in your eyes could cause eyestrain and headaches.

   b) What potential problem (strain/injury) could result if the window was directly behind you?

If a light source (eg window) is directly behind you, it can cause glare on your screen. This in turn can result in eyestrain and headaches.

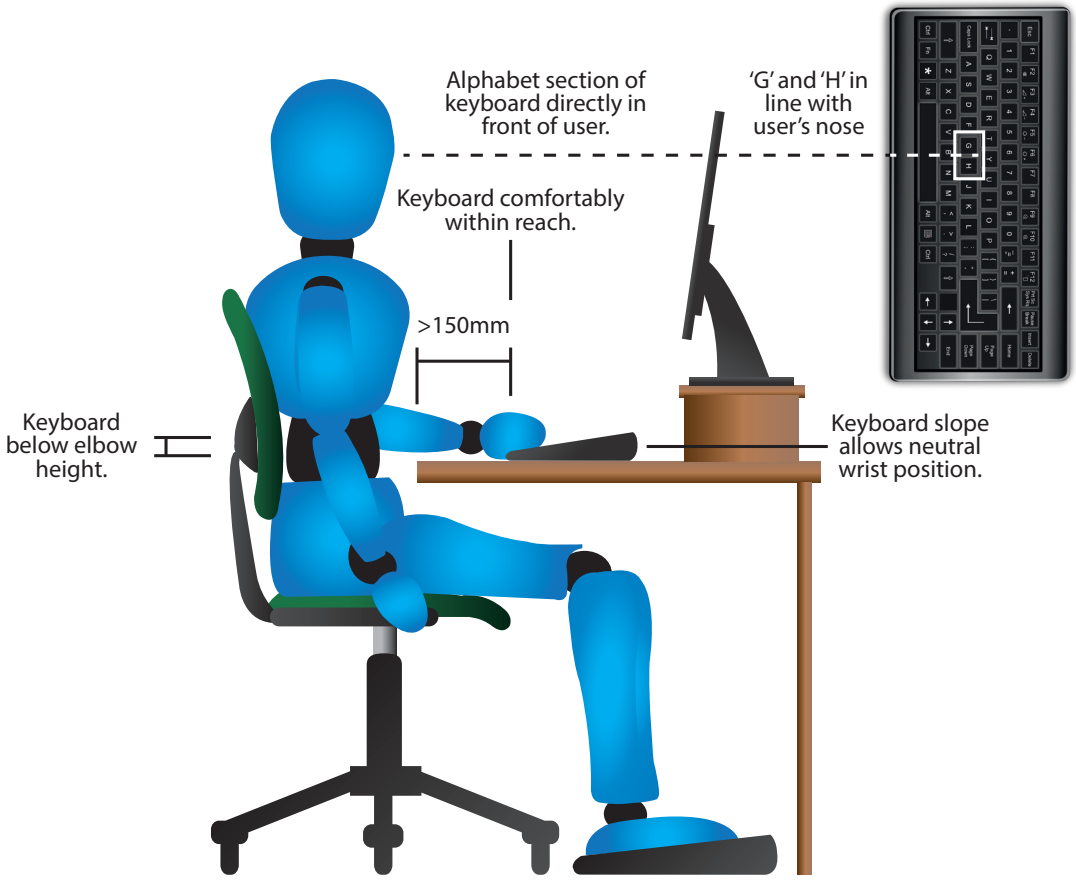
Keyboard and mouse

The mouse and keyboard need to be positioned in such a way that using them doesn't cause strain on the wrists, fingers, neck, shoulders, etc.

- The keyboard should be in line with the user's body - so that they don't have to twist in an awkward way to type.
- The keyboard and mouse must be close enough so that the user doesn't need to stretch to reach them. If the keyboard and/or mouse are too far away, it means that the muscles in your shoulder have to work harder to keep your arm and hand in position.

Position of keyboard

The diagram below shows the recommended positioning of the keyboard in relation to the user and computer hardware.



- Notice the diagram indicates that the keyboard's space bar needs to be at least 150mm from the edge of the table. This is to allow enough space for the user to rest their hands and forearms when they are not typing on the keyboard.

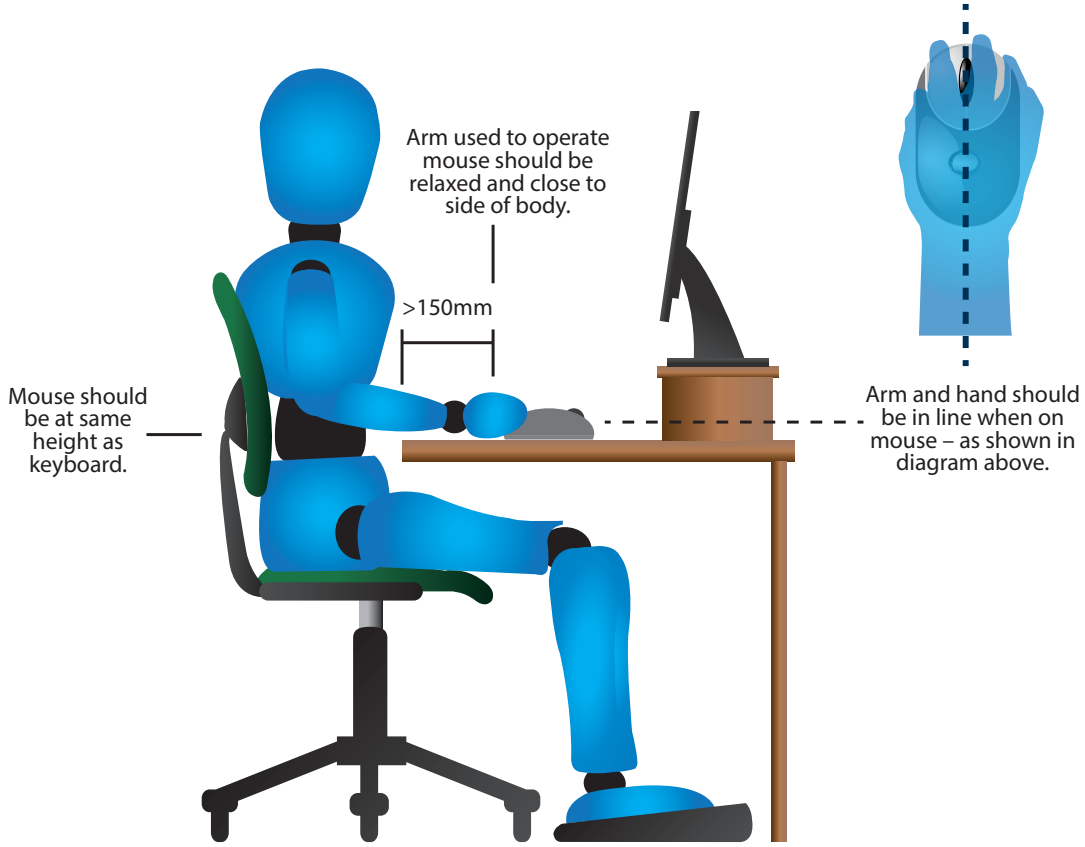
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NOTE: Remember that these are guidelines only, and are based on a person of average 'size' (height, weight, etc). For people who are much smaller or larger than average, the measurements indicated might need to be adjusted. For example, if have exceptionally long arms, 150mm might not be enough space for you to comfortably rest your hands and forearms.

- The main thing to remember about the position of the keyboard is that the person should be able use the keyboard without unnecessary strain on their hands, fingers, wrists, neck, back, etc.
- The keyboard needs to be stable - so that it doesn't move around when it is being used.

Position of mouse

The diagram below shows the recommended positioning of the mouse in relation to the user and computer hardware.



As is the case for a keyboard, the indication that the mouse should be at least 150mm away from the edge of the table is only a guideline. More or less space may be needed - depending on the size of the user (and in particular, their arm length).

General tips for mouse and keyboard use

When you use a mouse and/or keyboard you need to try to keep the muscles in your fingers, wrists, and arms as relaxed as possible.

- Avoid 'banging' on the keys of the keyboard (ie pushing really hard on the keys). This can lead to muscle tiredness and soreness.
- Avoid bending or twisting your wrist when typing on the keyboard or using the mouse.
- Make sure that when you are not typing or using the mouse, you relax your hand and wrist.
- Avoid gripping the mouse tightly or holding onto it when you're not using it. This can cause tension in your hand, wrist, and forearm.
- If possible, use keyboard shortcuts - to reduce the amount of work you need to do using the mouse.
- If possible, try using the mouse with your other hand. In other words, if you normally use the mouse with your right hand, try swapping to the left hand for a bit.

Healthy work habits

An important aspect of safe work practices in a digital environment is to take regular rest periods and exercise as part of your standard work practice. This helps to guard against physical discomfort and strain, physical and mental fatigue, and stress.

There are different ways to take a break from computer work and reduce the risk of problems.

Regular task breaks

- During a task break, you must not use the computer at all (not even for a 'relaxing' purpose). You need to leave the computer and do a different task, walk around, do a few stretches, etc.
- You should have these breaks regularly throughout the day.



Activity

1. Do you think it is important to take regular breaks when using a computer? If so, how often and how long do you think these breaks should be?

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It is important to take your tea and lunch breaks in order to ensure you take a regular break.

Micropauses

- These are shorter 'breaks' which you take at your computer.
- They should be built into the natural flow of your work.
- A micropause is a chance for you to give your muscles a chance to relax and to change your body and eye position.
- Examples of what you could do during a micropause:
    - Take your hands away from the keyboard/mouse and relax your arms down at your sides.
    - Completely relax your shoulders, arms, hands and fingers for a few seconds.
    - Change your eye position by, for example, looking away from the screen at something a long way away (at least 2-3 metres away). This helps prevent eye strain.

What can you do?

If you are a worker or learner in an organisation, you may not be able to apply all of the procedures and solutions discussed above. Instead, if you become aware of any issues, you should let an appropriate person know as soon as possible. An appropriate person might be your line manager, team leader, supervisor, or the organisation's health and safety officer.

If you feel you need extra support (eg training) to help you implement any of the procedures discussed above, you should also talk to an appropriate person to get this support.

As soon as you become aware of any discomfort, it is important that you act to identify the cause and deal with it. This will give you the best chance of dealing with the problem effectively. Also, it will stop the problem becoming more serious and turning into pain or even an injury.



Activity

Who should you report any potential health and safety problems to in your workplace or place of learning?

Even for those procedures and solutions you yourself cannot apply within an organisation, you can still apply them when using digital tools in your own home. Remember that these principles do not only apply to places of work or study, but **anywhere** that you use digital tools.

Lesson Review

Complete the exercises which follow to review what you have learned in this lesson.

1. Look at the diagrams below which show different ways in which a mouse can be held and used.

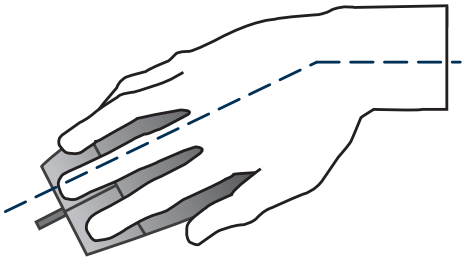


Diagram A

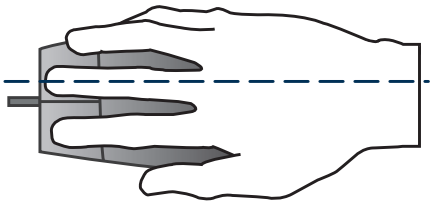


Diagram B

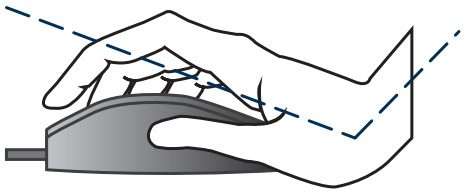


Diagram C

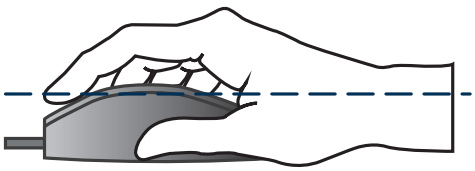


Diagram D

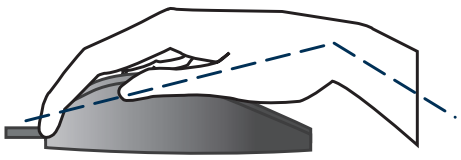


Diagram E

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1.1 Which diagram(s) show good ergonomic use of the mouse? Give reasons for your answer.

Diagrams B and D. In B, the mouse and the user's hand and wrist are in a straight line. This means that they are not straining their wrist to use the mouse. In D, the user's wrist is straight and parallel with the mouse. The wrist is not bent at an awkward angle, so this helps prevent strain, discomfort or injury of the wrist. In both diagrams, the user's hand looks relaxed on the mouse and they don't have to stretch or strain their fingers to use any part of the mouse.

1.2 Which diagram(s) show poor ergonomic use of the mouse? Give reasons for your answer and indicate what health problem could result from this.

Diagrams A, C and E. In all three of these diagrams, the user's wrist is bent at an awkward angle. This will cause pressure on the joints of the wrist and could also cause muscle strain in the forearm. The fingers are also awkwardly positioned (especially in Diagram C and E) and this could lead to strain in the fingers and hand.

2. Look at each of the diagrams below and on the next page. Then answer the questions that follow.



Diagram A



Diagram B

Continued on next page...



Diagram C



Diagram D

2.2 Which diagram(s) show a good, ergonomic user position/posture? Give reasons for your answer.

Diagram C. The user's body is in a comfortable, neutral position – with their joints lined up in a natural way. They can easily reach the keyboard and there is enough space for them to rest their wrist and forearm on the desk. The user's arms can comfortably hang at the side when they are not typing/using the mouse. There appears to be good support for their back and their feet are firmly on the footrest.

2.3 Which diagram(s) show a poor ergonomic user position/posture? Give reasons for your answer and indicate what health problems could result from this poor posture.

Diagrams A, B and D. In Diagrams B and D, it looks as if the guy is resting or having a break – which is good. However, he is doing so in a way that could still cause problems for him. In Diagram B the user in sitting in a way that could cause strain on his lower back. Also the keyboard is too close and because it is on his lap, his wrist will be at an awkward angle when he types.

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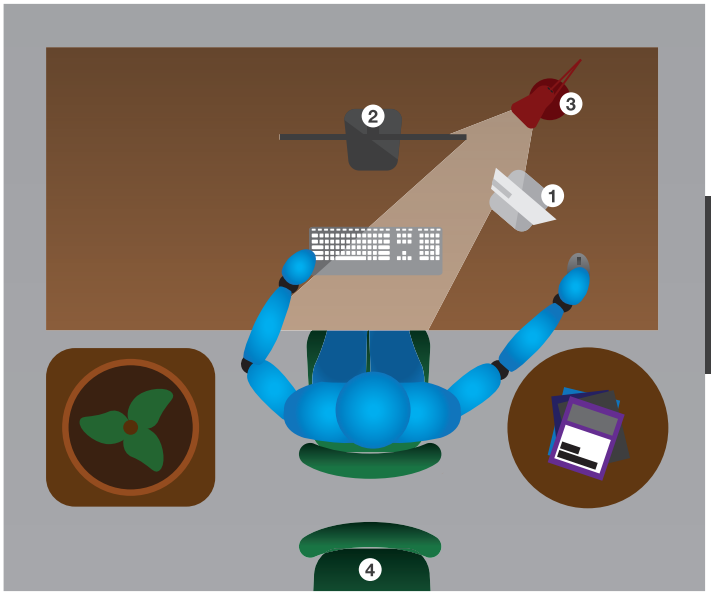
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The computer screen is also too far away for him to be able to see it without straining his eyes (unless it is a huge screen). If he stays in this position for quite a while, it is also likely to cause strain on the muscles behind his knees and on his ankles. This position also looks quite dangerous. He is right on the edge of his seat and looks as if he could easily fall off.

In Diagram D, it looks as if the user is taking a bit of a nap! However, as for Diagram B, his back is at an awkward angle and this is likely to cause lower back pain. Also, his neck is at a very awkward angle. He will soon get pain in his neck if he stays in this position.

In Diagram A, the main problem is that the user is too far from the desk. This means he has to stretch to reach the keyboard (and mouse?) and this could cause muscle strain and fatigue – especially in his arms, shoulders and wrists. Because of the way he is sitting in his seat, he is also not getting full support for his back.

3. The diagram below shows Aroha's workstation. Answer the questions on the next page.



Continued on next page...

3.1 What are some of the problems with the way things are arranged in Aroha's workstation?

3.2 Why is this a problem? In other words, what health problems might be caused or made worse by the way the workstation is arranged, and why?

3.3 What would you do to deal with the problem?

- 1. The mouse is too far away for her to use comfortably. This means she has to stretch to reach and work with the mouse. This could cause fatigue, strain and pain in her upper body, arm and hand. I would put the mouse within easy reaching distance for her and somewhere that she can use it without having to bend her wrist at an awkward angle.
- The desk lamp is positioned in such a way that (a) it shines in her eyes and (b) it casts shadows. This could lead to eye strain and pain, and possibly also headaches. I would move the desk lamp to the other side of the desk and angle it away from her face.
- She is 'boxed in' at her desk. This makes it difficult for her to move around regularly during the day and go to and from her desk with ease. I would move at least one of the side tables (on her left or right) away completely.

Teacher's Note:

It is not possible to cover all health risks and safety issues, and the procedures to address these, in the context of this guide. Therefore, learners should be encouraged to consult the ACC document **Guidelines for Using Computers - Preventing and managing discomfort, pain and injury**, for more detailed information. A copy of the guide is provided in the **Guide Resources** folder. Please note that the guide is copyright to WorkSafe New Zealand (WorkSafe) under the Creative Commons Attribution-Non-commercial 3.0 NZ licence.

Learners must also consult their organisation's policies and procedures document(s) related to health and safey, as individual organisations may have specific requirements in this regard.

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4. Read the cases studies that follow. These are all of people who are experiencing a health problem(s) linked to computer use.

   For each case study, indicate what you think:
   - the possible cause(s) of the problem is
   - the health problem(s) and their symptoms are
   - could be done to prevent or relieve the problem(s).

4.1 Maia is a secretary who spends a lot of time working at the computer. Unfortunately, her desk and chair are a little high for her.



Possible cause(s)	Incorrect sitting position caused by furniture that does not suit the user. Maia is straining to reach her keyboard (and other equipment) and because she is lower down than her desk, her wrist and arm are also at an awkward angle when she types or uses her mouse. Maia' feet can't rest comfortably on the ground.
Health problem(s) and symptoms	The health problems could be physical discomfort, pain or injury, or physical fatigue. Symptoms might include: • pain and stiffness in her wrists and hand • muscles in back, legs, and feet tired and stiff at the end of the work day.
Steps to prevent/ relieve the problem	If possible, Maia needs to adjust her chair and/or desk height so that she can be in a comfortable working position. • Her feet should rest comfortably on the ground. • She should be able to comfortably reach her mouse and keyboard without straining, and should be able to keep her wrists straight when using the mouse/keyboard. If her chair height can't be adjusted, a footrest could help Maia sit comfortably with her feet supported. If none of the above is possible, then Maia needs to speak to someone in charge, and make them aware of the problem.

Continued on next page...

4.2 Andrew loves playing games online. He plays role-playing games in which he interacts with players from around the world.

Every day when he gets home from school he plays online until the early hours of the next morning.

He has good posture when he plays online, and is sitting in a comfortable chair. However, he still finds he feels tired and has problems with his eyes.

Possible cause(s)	Very long periods in front of a computer and looking at a computer screen. Concentrating intensely for many hours while playing the game. Lighting could be too bright or screen could be reflecting too much glare.
Health problem(s) and symptoms	Physical and mental fatigue – he is overdoing it with long hours on the computer, after a long day at school. He is probably not getting enough sleep and concentrating intensely for long periods. He might find it difficult to do other tasks as a result of his 'over-activity' in playing on the computer. Visual discomfort – he might have sore, red, dry or watery eyes. His eyes could feel gritty or heavy. He might have blurred vision and as a result he could even end up getting headaches or migraines. Despite having good posture he might find that he gets discomfort, pain and fatigue in his muscles and joints from sitting in the same position for a long time, prolonged use of mouse, keyboard, joystick, etc.
Steps to prevent/ relieve the problem	He needs to cut down on the number of hours he plays online so that he gets enough sleep. When he is playing online, it is important that he takes regular breaks and micro-pauses. If necessary, he should adjust the brightness of his screen or add more light in the room – so that there isn't a sharp contrast between the two. He needs to make sure that he doesn't have any glare on his screen (eg from a lamp on his desk, etc).

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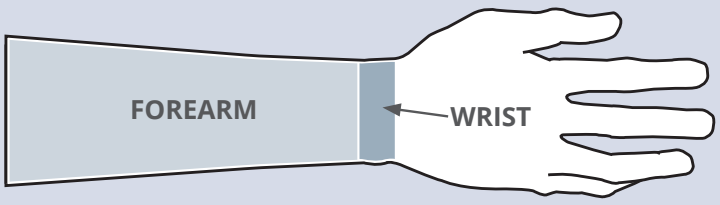
4.3 Sally sits for long periods talking to people on the phone and working in front of her computer.

She often needs to type and talk on the phone at the same time and so she squeezes the phone between her shoulder and ear to hold it in place.



Possible cause(s)	Holding the phone to her ear using her shoulder – for long periods of time. This causes her to use her shoulder, back and neck muscles to hold the phone in place for long periods.
Health problem(s) and symptoms	Physical fatigue – especially in shoulder, back and neck muscles. Possibly physical pain in the shoulder she uses to hold the phone, as well as her neck/back. She might also suffer from mental fatigue because she is doing two things at once for long periods of time.
Steps to prevent/ relieve the problem	Use a headset when needing to talk and type at the same time. Take regular breaks away from the computer (and telephone).

Glossary

absenteeism	Not coming to work.
adjustable	Can be changed.
compensate	Give someone something to make up for their suffering, loss, or something else bad that has happened.
compliance requirements	Law, rule, or standard that must be met.
confidentiality notice	Statement which indicates that material is confidential and what must be done if the material is sent to the wrong person by mistake.
consent	Agree to let someone do something.
contrast	Difference
data breach	When an unauthorised person gets access to personal information, either intentionally or by mistake.
deliberate(ly)	On purpose (not by mistake/accident).
digital environment	A situation or place where digital tools are used. This includes both hardware (digital devices, including storage and display device), software (applications), and data in digital form.
disclaimer	Statement used to limit responsibility.
discomfort	not feeling comfortable, feeling an ache, tenderness, etc
dormant	Asleep; inactive
employee turnover	People resigning from work.
ergonomics	The arrangement and use of furniture and digital device components in a way that helps to prevent and manage discomfort, pain, and injury.
fatigue	Tiredness
financial settlement	Pay someone money to make up for something you have done wrong.
firewall	Software used to protect a computer system against threats from the Internet.
forearm	
glare	Bright light that makes it difficult to see
hack	To access a computer and/or network without permission.

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implement	Put into practice.
injury	A physical wound or damage to the body, sometimes needing medical attention
integrity	For data to have integrity it should not have been changed, lost or destroyed.
intentional	Done on purpose (**not** by mistake or accident).
keystrokes	Each time you press a key it is a keystroke.
legal sanction	A form of punishment used by the law (eg a fine, being jailed, etc).
legitimate	Genuine, real. 'Legitimate computer programs' refers to software intentionally installed on a computer that is performing an expected task/function.
malicious	Intended to do harm.
malware	Short for 'malicious software'. This is software that is intended to do harm (eg a virus).
monotonous	The same thing over and over again.
organisation	Business, place of learning, etc.
outgoing email	Email that is being sent
personal information	Anything that can identify a person. For example, a photograph of the person, their name, address, etc.
phishing scams	Phishing (pronounced 'fishing') is a way people try to get sensitive information such as your username, password, credit card details, etc.
postural (muscles)	Muscles used to position your body.
postures	How you position your body.
psychological	To do with how you feel emotionally and mentally.
replicate	Make a copy of
self-replicating	Automatically makes copies of itself.
spam	An electronic message you did not ask for.
stable	Firm, solid, strong, steady
unauthorised	Not given permission. An unauthorised person is someone who has not been given permission to access or do something.
unintentional(ly)	Not on purpose; by mistake.
unsolicited	Not asked for/requested.
ventilated	Getting air to come into a room/building and move around the area freely.

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instant LTD
education solutions

21 Ruakura Rd, Hamilton East,
Hamilton 3216
Phone: 0800 864 863
E-mail: orders@instant.org.nz

www.instant.org.nz

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